TAFP-GraphRAG: An Auditable Framework for Sensitive-Relation Protection in Graph Retrieval-Augmented Generation

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Abstract

Graph Retrieval-Augmented Generation (GraphRAG) is an emerging intelligent retrieval architecture in which graph-structured evidence supports multi-hop reasoning and relation-aware context construction. However, relation labels and graph topology may reveal policy-restricted associations even when node identities are masked. This study presents Topology-Aware Fine-grained Protection for GraphRAG (TAFP-GraphRAG), an auditable pre-indexing framework for sensitive-relation protection in privacy-aware intelligent retrieval systems. The framework integrates policy-conditioned entity pseudonymization, concealment of designated sensitive-relation labels, constrained edge rewiring, exact per-node degree preservation in designated variants, and hash-traceable execution evidence. Across five dataset-derived graphs, the full configuration concealed 0.99993–1.00000 of designated sensitive relation labels while retaining 0.89935–0.92193 of non-sensitive relation semantics. All designated degree-preserving conditions preserved every node degree exactly. Under predefined retrieval-grounded structured relation-sequence question answering with deterministic decoding, TAFP-GraphRAG retained 93.8% of baseline non-sensitive relation-token F1 and reduced sensitive-relation disclosure by 92.2%. Across the full evaluation and within each dataset, it also achieved higher non-sensitive relation-token F1 and lower sensitive-relation disclosure than both evaluated sensitivity-blind controls. Attack outcomes remained dataset- and attacker-dependent: matched random perturbation

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remained competitive in several inference settings, original-edge existence could remain inferable, and the default protected marker did not conceal policy membership. These results support TAFP-GraphRAG as a reproducible and auditable framework for selective relation-semantic protection under the evaluated protocols; they do not establish a formal or universal privacy guarantee.

Keywords: Graph retrieval-augmented generation, Privacy-aware intelligent systems, Sensitive-relation protection, Graph perturbation,
Relation-inference attacks, Privacy–utility evaluation

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1. Introduction

Retrieval-Augmented Generation (RAG) improves the factual grounding of large language models by drawing on external evidence at inference time. Graph Retrieval-Augmented Generation (GraphRAG) extends this approach by representing documents, entities, concepts, and their relations as connected graphs. These structures support neighborhood retrieval, multi-hop traversal, and relation-aware context construction [1–8]. They also introduce a distinct information-security challenge. Sensitive information may reside not only in node attributes or source text, but also in relation labels, edge patterns, and local topology [9–16].

Conventional de-identification methods mainly remove or pseudonymize explicit identifiers. In graph-based systems, however, masking node identities does not necessarily conceal the associations between them. Relations such as diagnosis, treatment, employment, affiliation, transaction, or communication may remain visible after entity pseudonymization. Concealing original relation labels may still be insufficient: neighboring relation distributions, endpoint types, path structures, and other released local features can support relation inference [9, 10, 14, 15, 17, 18]. Sensitive associations may therefore remain inferable through explicit relation semantics, original-edge evidence, topology-based inference, learned graph representations, or relational context supplied to a language model [19–26].

Graph-privacy research provides important foundations, including node anonymization, edge deletion, randomized response, differential privacy, degree-preserving rewiring, synthetic graph generation, and attack-based auditing [17–19, 22, 25, 27–36]. Many existing transformations, however, do not distinguish relations that require protection from those that may remain available for retrieval. Uniform deletion or rewiring can reduce exposure while also removing useful non-sensitive evidence. Preserving aggregate graph statistics or node degrees does not by itself provide semantic selectivity. Previous work has examined anonymous-entity retrieval, privacy-aware graph prompting, graph-structure attacks, and privacy-preserving graph perturbation as separate problems. In the reviewed literature, we identified no GraphRAG framework that jointly integrates sensitivity-aware transformation of relation semantics, exact per-node degree preservation, auditable transformation records, and attack-specific privacy–utility evaluation.

We propose Topology-Aware Fine-grained Protection for GraphRAG (TAFP-GraphRAG), an offline transformation framework applied before indexing and retrieval. An explicit policy identifies the relations that require protection. The framework pseudonymizes policy-selected entities, conceals the original labels of designated sensitive relations, and rewires selected edges under predefined constraints. The transformation targets policy-designated relations while retaining non-sensitive relation semantics whenever the policy permits. In designated degree-preserving variants, every node retains its original degree exactly.

Indexing and retrieval operate on the protected graph, and the generator receives protected graph-derived context. Transformation-audit records remain with the graph owner and are excluded from the released adversarial view. Reproducible seeds, transformation summaries, integrity checks, runtime records, exact runner identities, and hash-verified evidence manifests support auditable execution.

The study evaluates whether policy-guided transformation can conceal designated relation semantics while retaining non-sensitive relational utility under explicit attack surfaces and structural constraints. The evaluation uses five dataset-derived graphs and a control hierarchy that separates sensitivity guidance from perturbation volume and structural preservation. The protocols cover semantic selectivity, structural invariants, original-edge inference, structure-only and embedding-based relation inference, release-marker leakage, policy misclassification, retrieval-grounded structural utility, and end-to-end relation-question answering. Paired multi-seed comparisons, exact executable-source identities, and hash-verified evidence records support reproducibility.

This study makes four contributions:

1. **Policy-conditioned selective relation protection.** We introduce a GraphRAG preprocessing framework that combines selective entity

pseudonymization, concealment of designated sensitive-relation labels, and constrained edge rewiring. The transformation allocates protection according to an explicit sensitivity policy rather than perturbing relation semantics uniformly.

2. **Verified structural invariants and auditable execution.** The evaluation verifies exact per-node degree preservation in every designated degree-preserving condition and distinguishes this invariant from stronger 2K joint-degree preservation. Exact runner identities, transformation audits, integrity checks, runtime records, and hash-verified artifacts make the implementation traceable.

3. **Controls and attacks that separate distinct security questions.** Deletion-only, perturbation-budget-matched random, and sensitivity-blind 2K controls separate semantic guidance from perturbation volume and structural preservation. Separate protocols evaluate original-label concealment, edge-existence evidence, policy-membership leakage, policy error, topology-only inference, spectral inference, and a released-view GCN-GAE stress test.

4. **Task-grounded privacy–utility evidence.** Structural retrieval metrics are reported separately from model-facing relation-question-answering outcomes. This distinction shows where sensitivity guidance does and does not provide an advantage: it does not generally maximize structural overlap, but under the predefined downstream protocol it preserves more non-sensitive answering utility while disclosing fewer sensitive relations than the evaluated sensitivity-blind controls.

2. Related Work

2.1. GraphRAG and Relation-Level Exposure

Graph Retrieval-Augmented Generation (GraphRAG) represents retrieval evidence as connected structures of documents, entities, concepts, and relations. Its main stages typically include graph construction, indexing, structured retrieval, and graph-enhanced generation [5]. GRAG retrieves textual subgraphs to organize evidence [1], whereas topology-aware methods use graph connectivity directly during evidence selection [6]. More recent systems combine graph traversal, retrieval planning, hierarchical organization, and multi-level pruning to support multi-hop reasoning and reduce irrelevant context [3, 4, 8]. Medical Graph RAG and MedSumGraph adapt this architecture to clinical retrieval and question answering [2, 7].

Because relation labels and local topology become part of the evidence supplied to retrieval and generation, GraphRAG can expose information beyond explicit entity identities. Pseudonymization may leave sensitive associations visible, and concealed relation labels may remain inferable from neighborhood structure or other released local features. These observations motivate evaluating explicit label concealment separately from inference based on released structural cues [9, 10, 14, 15, 17, 18, 37].

GraphRAG-specific security research now examines this exposure directly. One recent study reports the extraction of structured entity–relation information as well as raw text through black-box queries [37]. AGEA studies budget-constrained adaptive extraction of a hidden entity–relation graph [38]; GRASP examines targeted multi-turn subgraph reconstruction against deployments protected by prompt-level safeguards [39]; and GraphSteal reconstructs hidden structure through adaptive traversal of the system as a structural oracle [40]. GraphRAG under Fire addresses a different security objective by examining poisoning attacks that exploit graph-based indexing and retrieval [41].

Selective-disclosure work addresses another layer of the problem. SD-RAG applies dynamic, policy-aware sanitization during retrieval through a graph-based data model [42]. Query-driven extraction, runtime disclosure control, and pre-indexing graph transformation are related but distinct security layers. Adaptive multi-turn reconstruction targets a broader system-extraction surface than the fixed released-view and relation-inference protocols considered here, so these threat models are complementary rather than equivalent.

2.2. Privacy in Retrieval and Graph-Based Systems

Privacy research on retrieval-augmented systems examines different information objects, protection mechanisms, and attack objectives. Anonymous-entity retrieval removes explicit knowledge-graph identities [24]; locally private entity perturbation protects entity-level information in conventional RAG [21]; and synthetic-corpus methods replace or regenerate source records [26]. Membership-inference studies instead ask whether a particular record belongs to the retrieval collection [20, 22].

These approaches address important risks, but their reported objectives and evaluations do not by themselves establish protection against relation-semantic exposure. Removing an entity name may leave the associated relation type visible, while document-membership protection does not establish whether graph context reveals a sensitive relation label or allows that label to be inferred from neighboring structure.

Graph-focused studies describe a broader attack surface. Privacy-aware graph prompting has been investigated through adaptive perturbation and topology-recovery mechanisms [23]. Other work reports risks from graph prompting, link stealing, graph-property inference, topology inference, and practical edge-inference attacks [9, 10, 14–18]. Research on graph databases, federated knowledge-graph embeddings, and vertical federated learning further shows that query outputs, shared representations, and intermediate computations may expose entities, triples, adjacency information, or relation labels [11–13].

These findings concern different security outcomes: recovery of an original relation label, inference of original-edge existence, topology-mediated relation classification, and disclosure through model-facing context. They should not be collapsed into a single leakage measure, and relation-label concealment should not be treated as complete graph anonymization.

2.3. Graph Perturbation, Differential Privacy, and Structural Constraints

Differentially private graph learning, private graph publication, private graph generation, and locally private edge or relation perturbation provide formal guarantees under defined neighboring-graph relations, trust models, and privacy budgets [28–30, 32, 34, 36]. These methods are appropriate when a deployment requires a quantified formal privacy guarantee.

TAFP-GraphRAG addresses a complementary systems objective. It transforms an existing GraphRAG graph according to an explicit sensitivity policy, preserves specified structural constraints, and seeks to retain non-sensitive relation semantics. Exact per-node degree preservation in designated variants is a deterministic structural invariant, not a differential privacy guarantee.

Direct numerical comparison across protection layers requires alignment of the protected information object, adversarial interface, released representation, and privacy or perturbation budget. Without such alignment, runtime disclosure controls, extraction defenses, formal graph-privacy mechanisms, and pre-indexing relation or topology transformations answer different security questions and should not be treated as interchangeable baselines.

TAFP-GraphRAG is therefore evaluated against controls aligned with its transformation mechanism and structural invariants. Degree-preserving double-edge switching replaces two edges while preserving the degree of each participating node [31]. Deletion-only dropout, perturbation-budget-matched random controls, and a sensitivity-blind 2K baseline that preserves individual node degrees and the complete undirected joint-degree distribution distinguish modification quantity, structural constraints, and the sensitivity policy that determines where transformation is applied [33, 35].

2.4. Privacy–Utility Evaluation and the Remaining Gap

Graph anonymization is commonly evaluated through aggregate graph statistics, downstream prediction tasks, or attack-specific privacy measures. DUEF-GA provides a general framework for examining graph privacy and utility together [27]. Empirical privacy auditing additionally emphasizes explicit adversarial observations, reproducible attack construction, and measurable inference advantage [19, 25].

GraphRAG adds a task-specific utility requirement: a transformed graph must continue to support useful evidence retrieval even when paths, neighborhoods, or relation labels change. Global graph similarity alone cannot determine whether predefined query endpoints remain reachable, relevant subgraphs are retained, or non-sensitive relational evidence remains available to the generator.

The gap addressed here is integrative rather than primitive-level. In the reviewed literature, we identified no prior GraphRAG framework that jointly integrates sensitivity-aware perturbation of relation semantics, exact per-node degree preservation, auditable transformation logging, and attack-specific evaluation of privacy–utility trade-offs. TAFP-GraphRAG addresses this gap by coupling pre-indexing graph transformation with attack-specific privacy–utility evaluation in a unified framework. Its novelty does not depend on presenting pseudonymization or edge switching as new primitives. Instead, it coordinates sensitivity guidance, exact structural constraints, auditable execution, and attack-oriented evaluation. Conventional structural constraints determine how a graph may be modified; TAFP-GraphRAG also uses an explicit disclosure policy to determine where those modifications are concentrated. The objective is selective concealment of original sensitive-relation labels while retaining non-sensitive relational utility under the evaluated conditions.

3. System and Threat Model

3.1. System Setting and Formalization

Each dataset-derived graph is modeled as a simple undirected attributed graph

$$G = (V, E, X, R), \tag{1}$$

where V is the node set, E is the undirected edge set, X denotes node and edge attributes, and R denotes exposed relation labels. Dataset-specific adapters construct G before protection. Across experimental conditions, the graph-construction rules, node and edge ordering, evaluation candidates, and query definitions remain fixed.

TAFP-GraphRAG applies an offline transformation before indexing and retrieval. Given transformation parameters θ and protection seed z, its interface is

$$(G', A) = T_{\theta,z}(G), \tag{2}$$

where G' is the protected graph and A is the implementation-level transformation-audit record. Only policy-selected entities are pseudonymized, and only policy-designated sensitive relation labels are concealed. Original labels of non-selected entities and non-sensitive relations may remain available when permitted by the policy.

Indexing and retrieval operate on G', and model-facing graph-derived context is constructed from G' rather than from the original graph. The audit record A remains with the graph owner and is excluded from the released adversarial view.

Figure 1 summarizes the offline transformation and downstream GraphRAG workflow.

3.2. Protection and Utility Objectives

The evaluation addresses five distinct objectives: conceal the exposed labels of policy-designated sensitive relations; retain non-sensitive relation semantics when the policy permits; preserve every node degree exactly in designated degree-preserving variants; retain sufficient path, neighborhood, and relation evidence for the predefined retrieval and relation-question-answering tasks; and maintain reproducible, auditable transformation behavior through fixed seeds, integrity checks, run-level records, exact executable-source identities, and hash-verified artifacts.

These objectives are not interchangeable. Exact degree preservation is a deterministic structural invariant, not a privacy guarantee. Endpoint reachability does not establish preservation of the original path, relation semantics, community structure, or other higher-order topology. Similarly,

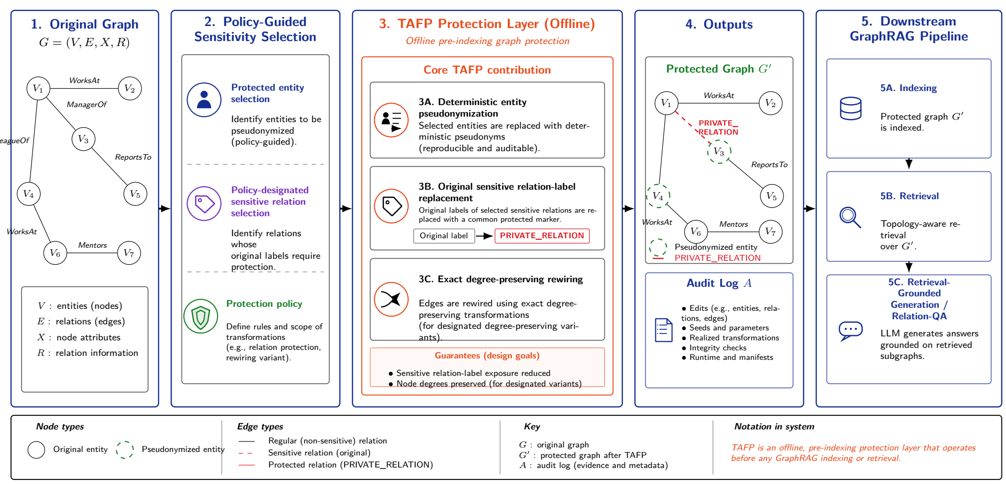

Figure 1: TAFP-GraphRAG offline protection and downstream workflow. A policy selects entities for pseudonymization and designates sensitive relations for protection. The offline transformation pseudonymizes selected entities, conceals the original labels of designated sensitive relations, and, in designated degree-preserving variants, rewires edges while preserving every node degree exactly. Indexing, retrieval, and retrieval-grounded relation question answering then use the protected graph. The transformation-audit record remains with the graph owner and is excluded from downstream context.

concealing an original sensitive relation label does not establish protection against inference of original-edge existence, policy membership, or relation class through another observation surface.

TAFP-GraphRAG is evaluated as an empirical, policy-conditioned protection mechanism rather than a formal privacy mechanism. It does not provide differential privacy or local differential privacy and does not claim complete anonymization, raw-text confidentiality, document-membership privacy, edge privacy, topology privacy, or universal protection against all attackers.

3.3. Adversary Model and Evaluation Scope

The adversary varies by protocol. Depending on the evaluation, it may observe the protected graph, protected node labels or pseudonyms, permitted entity types, protected relation labels, transformed connectivity, and protocol-defined labeled training examples. The released protected graph view excludes explicit sensitivity indicators, sensitivity scores, raw audit records, and the original labels of designated sensitive relations; original labels of non-sensitive relations may remain when the policy permits. Protocol-defined labeled training examples and evaluator-side ground truth are separate from the released protected graph view and follow the corresponding evaluation protocol.

Six primary exposure channels are evaluated: direct recovery of original relation labels; inference that a node pair was connected in the original graph; structure-only relation inference; spectral-embedding relation inference; released-view attribute-aware GCN-GAE relation inference; and disclosure of original sensitive relations through retrieval-grounded model outputs. A separate sensitivity-marker audit tests whether released protected-label patterns reveal membership in the supplied sensitive-policy set. Policy-misclassification experiments assess dependence on the supplied policy and are not treated as an additional adversarial observation channel.

The released-view and locked end-to-end protocols exclude raw source text. A separate controlled seed-42 model-facing protocol includes node-text fields truncated to 180 characters only to test whether a generator transmits a relation token already exposed in the transformed context. This controlled context is distinct from both the released-view attacker and the locked end-to-end protocol.

The study uses fixed, predefined attacks and query sets. Adaptive multi-turn extraction, unrestricted free-form graph reconstruction, coordination across multiple releases, continual-update attacks, and complete subgraph reconstruction remain outside the evaluated scope. Claims are therefore limited to the specified datasets, policies, models, released attributes, attack features, and observation surfaces.

Figure 2 summarizes the evaluated observation surfaces and protocol boundaries.

4. TAFP-GraphRAG Protection Framework

Building on the graph definition in Eq. (1) and the transformation interface in Eq. (2), TAFP-GraphRAG applies an offline, policy-guided composition of four ordered operations before GraphRAG indexing and retrieval:

$$T_{\theta,z} = W_T^{\rho_T} \circ W_R^{\rho_R} \circ S_E \circ P_V. \tag{3}$$

Here, P_V pseudonymizes policy-selected entity labels, S_E conceals the exposed labels of policy-designated sensitive relations, $W_R^{\rho_R}$ applies relation-level rewiring at rate ρ_R, and $W_T^{\rho_T}$ applies topology-level rewiring at rate ρ_T. Because the composition is read from right to left, relation-semantic concealment precedes both rewiring phases.

In the designated degree-preserving variants, the structural operations satisfy the first-order invariant

$$d_{G'}(v) = d_G(v),\ \forall v \in V. \tag{4}$$

This invariant concerns node degrees only and does not extend to connected components, paths, clustering, motifs, communities, spectral properties, or retrieval neighborhoods.

4.1. Policy-Guided Selection and Deterministic Pseudonymization

A node is eligible for policy-guided pseudonymization when its PII indicator is true or its semantic-sensitivity value is at least the policy threshold. Eligible nodes are ranked by $2 \times \text{PII} + \text{sensitivity}$, treating the Boolean PII indicator numerically; ties are resolved deterministically by a hash-derived value computed from the node identifier and protection seed z.

The default node-sensitivity threshold is 0.45 and the default entity-selection rate is 0.75. The selected-node target is computed with Python's built-in rounding applied to the eligible-candidate count multiplied by the selection rate. For a positive rate and a non-empty candidate set, the target is raised to at least one and capped at the candidate count.

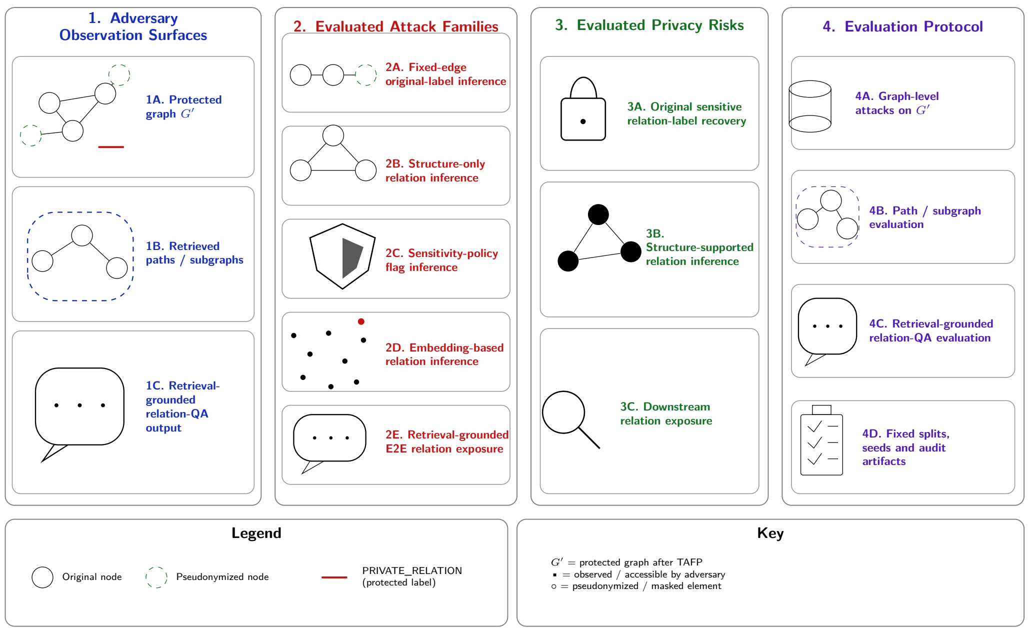

Figure 2: Threat model and evaluation scope. The adversary is assessed across three observation surfaces: the released protected graph, retrieved path or subgraph context, and retrieval-grounded relation-question-answering output. Primary analyses cover direct original-sensitive-label exposure, original-edge-existence inference, structure-only relation inference, spectral-embedding relation inference, released-view attribute-aware GCN-GAE relation inference, and end-to-end relation exposure. A separate marker-leakage audit tests whether released protected-label patterns reveal policy membership. The explicit internal sensitivity-policy flag and transformation-audit records are excluded from the released view.

Each selected entity receives an exposed alias of the form `ENT_<LEVEL>_<HASH>`. The level is HIGH when PII is true or sensitivity is at least 0.70, MED when PII is false and sensitivity is at least 0.45, and LOW otherwise. The suffix is the first eight hexadecimal characters of `md5(f"{node}-{z}")`. This suffix provides deterministic experimental aliases; it is not encryption, keyed hashing, or a claim of irreversible anonymization. The implementation contains no separate alias-collision detection or collision-resolution mechanism, and no global alias-uniqueness guarantee is made.

4.2. Relation-Semantic Concealment and Structural Transformation

The relation policy designates sensitive edges using an explicit internal indicator and/or a relation-sensitivity score, with a default score threshold of 0.45. These are operational policy inputs rather than calibrated probabilities or differential-privacy parameters. Before structural rewiring, the exposed protected relation label of each policy-designated sensitive edge is replaced by `PRIVATE_RELATION`. This replacement conceals the original relation type in the released label field, but the marker can reveal membership in the policy-protected set; original-label concealment and marker-membership concealment are therefore evaluated separately.

The relation phase operates on current edges that satisfy the internal sensitivity rule. The topology phase then operates on the complete current edge set produced by the relation phase.

For each accepted double-edge switch, the current attribute dictionary of the first source edge is copied to the first inserted edge, and the second source dictionary is copied to the second inserted edge. During relation-level rewiring, each copied dictionary is sanitized before insertion by setting `relation_type` and `protected_relation_type` to `PRIVATE_RELATION`, `sensitive` to 0, and `relation_sensitivity` to 0.0. During topology-level rewiring, the corresponding current source-edge dictionaries are transferred unchanged, without additional topology-phase sanitization. Attribute transfer is therefore order-specific: source dictionaries are neither randomly assigned to inserted edges nor merged.

4.3. Exact Per-Node Degree-Preserving Rewiring

For a phase with m candidate edges and rate ρ, the target rewired-edge count is zero when $m < 2$ or $\rho \leq 0$. Otherwise, the implementation computes `int(round(`$m \times \rho$`))`, raises the result to at least two, caps it at m, and subtracts one if the capped value is odd. The result is a target number of rewired edges, not accepted switches, because each accepted switch rewires two edges.

The proposal limit is $\max(2000, 300n)$, where n is the target rewired-edge count. If the limit is exhausted before n edges are rewired, execution completes without an exception and returns the target and realized counts separately; convergence to the target is not guaranteed.

Given source edges $\{u, v\}$ and $\{x, y\}$, a proposal is considered only if both source edges still exist and the four endpoints are distinct. A reconnection is accepted only if it creates no self-loop, the two inserted edges are distinct, and neither inserted edge already exists. Each accepted switch removes and adds one incident edge for every participating endpoint, thereby preserving every node degree in variants composed of these switches. This guarantee is restricted to the designated degree-preserving variants and does not extend to higher-order graph properties.

Figure 3 illustrates an accepted relation-phase switch and the resulting degree invariant.

4.4. Auditable Execution and Operating-Point Selection

The in-application `protection_audit` record stores implementation-level summaries rather than a complete event-by-event transformation history. Its verified fields are `sensitive_node_candidates`, `masked_nodes`, `entity_mask_rate`, `sensitive_edges`, `relation_rewired_edges`, `relation_target_edges`, `topology_rewired_edges`, `topology_target_edges`, `degree_changed_nodes`, and `degree_preserved_exactly`. The record is not described as containing complete removed-edge, added-edge, or pseudonymized-node lists. Protection seeds, runtime measurements, and artifact hashes are maintained in separate run-level tables and hash-verified manifests rather than being assumed to reside inside `protection_audit`.

The default parameter values are pre-specified operating points examined through the reported ablations; they were not selected through held-out deployment calibration. For deployment, an operator may define acceptable concealment, non-sensitive retention, and structural requirements on held-out policy-calibration data, select the least disruptive configuration that satisfies them, and confirm the selection on a separate audit set. This is an empirical configuration procedure, not formal privacy accounting.

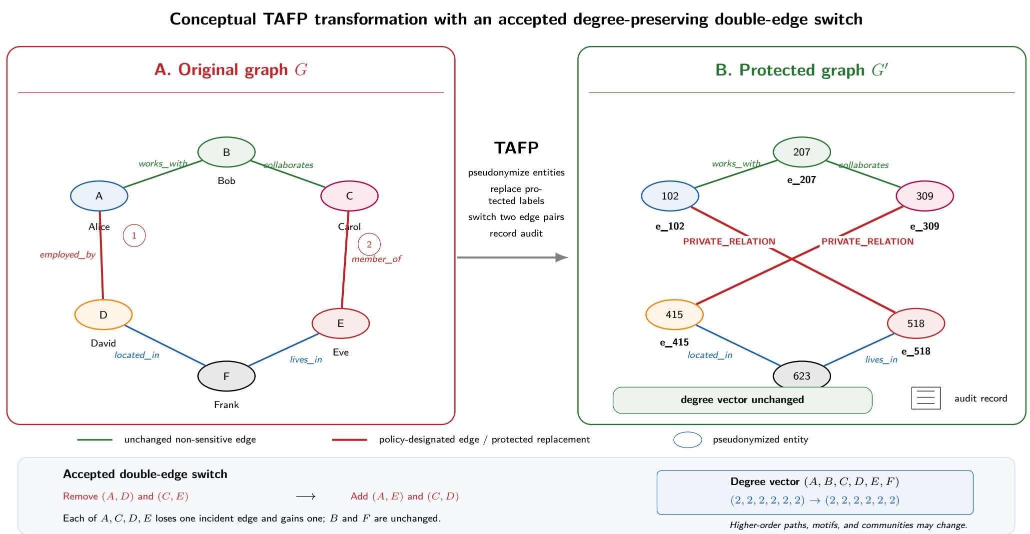

Figure 3: Conceptual relation-phase example of an accepted degree-preserving double-edge switch in TAFP-GraphRAG. The original sensitive edges (A,D) and (C,E) are removed and replaced by (A,E) and (C,D). In this illustration, both source relations have already been assigned the protected label PRIVATE_RELATION; therefore, both inserted edges are released with that protected label. Because each of A, C, D, and E loses one incident edge and gains one, every node degree remains unchanged. The example illustrates the relation-focused case and the degree invariant only; higher-order paths, motifs, and communities may change.

Algorithm 1. Policy-guided TAFP-GraphRAG transformation
Input: original simple undirected attributed graph G; protection policy π; transformation parameters θ, including the entity-selection rate, ρ_R, and ρ_T; protection seed z.
Output: protected graph G'; implementation-level transformation-audit record A.

1. Copy G to G' and initialize A.
2. Identify the policy-eligible entity candidates in G'.
3. Rank candidates by $2 \times \mathrm{PII} + \mathrm{sensitivity}$; resolve equal scores deterministically from the node identifier and z.
4. Compute the entity-selection target using Python rounding, the positive-rate minimum of one, and the candidate-count cap.
5. For each selected entity, assign HIGH, MED, or LOW and replace its exposed label with `ENT_<LEVEL>_<HASH>`.
6. Replace the exposed labels of policy-designated sensitive relations with `PRIVATE_RELATION`.
7. Relation phase: form the current sensitive-edge candidate set; compute n using the verified rounding, minimum, cap, and even-parity rules; set the proposal limit to $\max(2000, 300n)$; apply the switch-acceptance checks; copy first-source attributes to the first inserted edge and second-source attributes to the second; sanitize the copied relation attributes before insertion; increase the realized rewired-edge count by two for each accepted switch.
8. Topology phase: use the complete current edge set; compute the target and proposal limit with ρ_T; apply the same acceptance checks; transfer corresponding current source-edge attributes unchanged; increase the realized rewired-edge count by two for each accepted switch.
9. Compare the degree of every node in G' with its degree in G.
10. Record the verified implementation-level audit summaries in A.
11. Return G' and A.

5. Experimental Design and Evaluation Methodology

5.1. Dataset-Derived Graphs and Fixed Evaluation Protocol

The evaluation used five dataset-derived graphs: AGNews [43], CNN/DailyMail [44], IMDB [45], PubMed [46], and SynthPII, which was generated for this study. AGNews derives from a four-class news-classification corpus, IMDB from the Large Movie Review Dataset, CNN/DailyMail from article–summary pairs, and PubMed from biomedical articles and abstracts in a long-document summarization collection. SynthPII represents synthetic personally identifiable entities and sensitive workflow relations.

Dataset-specific adapters converted each source into a simple undirected attributed NetworkX graph. Depending on the dataset, nodes represented documents, summaries, reviews, topics, entities, biomedical concepts, or synthetic organizational records; edges represented typed semantic relationships. Self-loops and duplicate undirected edges were excluded during graph construction.

The resulting graphs contained 2,614 nodes and 6,542 edges for AGNews; 4,953 nodes and 15,248 edges for CNN/DailyMail; 3,119 nodes and 8,102 edges for IMDB; 1,070 nodes and 4,254 edges for PubMed; and 800 nodes and 2,400 edges for SynthPII.

Graph-construction rules, node and edge ordering, evaluation candidates, and query definitions were held constant across all protection variants and protection seeds. The structural-utility protocol comprised 2,208 fixed paths and 750 source–hop contexts defined before protection. The same source nodes, target nodes, original paths, and hop radii were reused across conditions. Table 1 reports the dataset and fixed-protocol characteristics.

Table 1: Dataset and experimental-protocol characteristics. Fixed paths and source-hop contexts were defined before protection and reused across all variants and seeds.

Dataset	Nodes	Edges	Fixed paths	Source-hop contexts
AGNews	2,614	6,542	438	150
CNN_DM	4,953	15,248	448	150
IMDB	3,119	8,102	436	150
PubMed	1,070	4,254	442	150
SynthPII	800	2,400	444	150

5.2. Sensitivity Policies and Experimental Operating Points

Each dataset-derived graph contains node- and edge-level policy attributes used to define the experimental protection scope. Under the defined selection rules, a node was eligible for pseudonymization when its PII indicator was true or its sensitivity score was at least 0.45. Eligible nodes were ranked by $2 \times \text{PII} + \text{sensitivity}$ with deterministic seed-dependent tie ordering. The default entity-selection rate was 0.75, but the selected-node target applied the implementation's integer-rounding and bounding rules to 0.75 times the eligible-candidate count and was not assumed to equal exactly 75% in every graph.

An edge was policy sensitive when its internal sensitivity indicator was true or its relation-sensitivity score was at least 0.45. The default relation- and topology-level rewiring rates were 0.12 and 0.08, respectively. These values were pre-specified operating points, not calibrated probabilities, privacy budgets, or formal risk scores.

In the final loaded graphs, policy-sensitive edges numbered 3,552 of 6,542 (54.30%) for AGNews, 13,996 of 15,248 (91.79%) for CNN/DailyMail, 5,680 of 8,102 (70.11%) for IMDB, 4,101 of 4,254 (96.40%) for PubMed, and 2,207 of 2,400 (91.96%) for SynthPII. These prevalences describe the original proxy policies used in the primary matrix.

A separate minority-sensitive ablation evaluated target prevalences of 5%, 10%, 20%, and 30%. For each dataset, the node file was copied unchanged; edge-row structure and relation information were retained; the original sensitivity indicator was copied to `original_sensitive`; and the active `sensitive` field was reset and reassigned deterministically. Rows were selected only from the original policy-positive candidate set using policy seed 20260621. The ablation therefore tests lower prevalence within the original policy-positive candidates rather than policy misclassification. Five datasets, four prevalence levels, 15 protection seeds, and four graph conditions produced 1,200 conditions. The PubMed source-row targets were 216, 433, 865, and 1,298 for the four prevalence levels.

Hyperparameter sensitivity was evaluated one factor at a time. Entity-selection rates were 0.50, 0.75, and 1.00; relation-level rewiring rates were 0.06, 0.12, and 0.18; and topology-level rewiring rates were 0.04, 0.08, and 0.12. All non-varied parameters remained at their default values, and the same 15 protection seeds were used for every setting.

5.3. Protection Variants and Matched Controls

The primary experimental matrix used 15 protection seeds: 1, 7, 21, 42, 99, 123, 202, 303, 404, 505, 606, 707, 808, 909, and 1001. Five datasets and seven primary variants produced 525 graph conditions.

`baseline` applied no pseudonymization, relation-label concealment, rewiring, or deletion. `relation_only` replaced sensitive relation labels with `PRIVATE_RELATION` and applied degree-preserving relation-level rewiring to sensitive-edge candidates, without entity pseudonymization or topology-level rewiring. `topology_only` applied degree-preserving topology-level rewiring over the complete current edge set, without entity pseudonymization or systematic sensitive-label replacement. `full` combined entity pseudonymization, sensitive-label concealment, relation-level rewiring, and topology-level rewiring.

Three controls separated sensitivity guidance from transformation quantity and structural preservation. `random_relation_budget` selected from the complete edge set the same number of edges as the policy-sensitive count and applied the same relation target rules as `relation_only`. `random_edge_dropout` removed an even number of uniformly sampled edges without adding replacements; it is an intentionally non-degree-preserving deletion reference, not a matched sensitivity-guidance control. `random_full_budget` randomly selected the same entity target count and edge budget as `full`, used the same relation and topology rates, and removed policy guidance.

The pre-specified semantic-selectivity comparisons were `relation_only` versus `random_relation_budget` and `full` versus `random_full_budget`. They test whether sensitivity guidance changes which relation semantics are concealed and retained when transformation quantity is matched.

A separate sensitivity-blind 2K control, internally named `dk2_full_budget`, used the same entity, relation, and topology targets as the full-budget conditions while preserving exact per-node degree and the complete undirected joint-degree distribution. The 2K experiment added 75 unique transformations across five datasets and 15 seeds. The three-way analysis of full, `random_full_budget`, and 2K contained 225 condition records; full and random-full records were reused from the primary matrix rather than regenerated.

The controls address different questions. Random edge dropout measures structural deletion, the matched random controls isolate policy guidance at comparable transformation budgets, and the 2K control tests whether stronger degree-distribution preservation alone reproduces TAFP-GraphRAG's selective semantic behavior. None is treated as a universal lower or upper bound on privacy or utility.

5.4. Security and Privacy Evaluation Protocols

Protection was evaluated across distinct adversarial observation surfaces rather than summarized by a single privacy score. The protocols separately examined original-edge existence, recovery of original relation types after explicit labels were excluded, supplied-policy membership inferred from released protection markers, robustness to incorrect sensitivity assignments, and relation inference from learned graph representations.

The fixed-edge protocol measured residual evidence that a node pair had been connected in the original graph. Sensitive and non-sensitive positive edges formed separate target groups. Each positive edge was paired with a unique negative pair absent from the original graph and matched by endpoint-degree classes. Five datasets, 15 protection seeds, seven primary variants, and two target groups produced 1,050 condition summaries and 210,000 candidate-level records. Outcomes were local structural AUC, hybrid edge-inference AUC, positive-edge retention, original relation-label recovery, conditional relation recovery among retained edges, and the negative false-edge rate.

The structure-only relation-inference protocol tested whether an edge's original relation type could be predicted from local topology after explicit relation labels and released node attributes had been excluded. PubMed and SynthPII were evaluated under ten stratified train–test partitions, 15 protection seeds, and seven primary variants, yielding 2,100 attack conditions. Endpoint-pair features included endpoint-degree summaries, common-neighbor measures, Jaccard and resource-allocation scores, endpoint clustering statistics, shortest-path distance after removal of the target edge, and an indicator for disconnected endpoint pairs. Entity types and neighboring released relation-label distributions were excluded. Feature dictionaries were vectorized, scaled without mean centering, and classified by L2-regularized multinomial logistic regression. Macro-F1 was primary and balanced accuracy secondary.

A separate sensitivity-marker protocol tested whether released protected-label patterns disclosed membership of an original edge in the supplied sensitive-policy set. Before transformation, each original edge received provenance fields for its original endpoint pair, original relation label, original policy-sensitivity class, and a deterministic cover-set score. Five datasets and 15 protection seeds were evaluated under seven release modes: the default private marker, relation-label omission, and shared coarse-cover settings applied to 10%, 25%, 50%, 75%, or 100% of non-sensitive relations, yielding 525 conditions. The protocol separately measured provenance-level policy-membership AUC, marker precision, provenance-level sensitive-relation concealment, endpoint-conditioned concealment, global non-sensitive semantic retention, original-edge retention, retained-endpoint collisions, and exact degree preservation. Each shared-cover mode was compared with the default marker using paired exact sign-flip tests, 20,000-replicate bootstrap confidence intervals for mean differences, and Holm adjustment within each metric.

Sensitivity-policy quality was assessed through a separate misclassification-robustness protocol. For each dataset and protection seed, deterministic edge-level scores perturbed the clean oracle policy. False-negative scenarios removed supplied-policy labels only from oracle-sensitive edges, false-positive scenarios added labels only to oracle-non-sensitive edges, and symmetric scenarios allowed both error types. Rates of 5%, 10%, and 20% were defined from common deterministic scores, making perturbation sets nested within each error family. Five datasets, 15 protection seeds, and ten policy scenarios produced 750 conditions. Supplied-policy fidelity was reported separately from oracle-grounded concealment and preservation so that faithful execution of an incorrect supplied policy was not interpreted as protection under the original policy. The protocol also audited exact degree and edge-count preservation. Each noisy-policy condition was compared with the clean-policy condition using paired exact sign-flip tests, 20,000-replicate bootstrap confidence intervals, and Holm adjustment within each metric.

Embedding-based relation inference used deterministic 16-dimensional spectral node representations. For each protected graph, the adjacency matrix was symmetrically normalized as $D^{-1/2}AD^{-1/2}$. The largest 17 eigenvectors were computed, the leading eigenvector was excluded, and the remaining 16 dimensions were row normalized. For each endpoint pair, the attack representation concatenated the absolute embedding difference, Hadamard product, vector sum, dot product, and Euclidean distance; relation labels were not input features. PubMed and SynthPII were evaluated over 15 protection seeds, ten edge-disjoint split seeds, five graph variants, and two attack models, yielding 3,000 model evaluations. The first attacker used standardized multinomial logistic regression with balanced class weights. The second used standardized features, deterministic class-balanced resampling of the training set, and a multilayer perceptron with hidden layers of 64 and 32 units. Macro-F1, balanced accuracy, and accuracy were recorded together with degree-preservation, edge-count, convergence, and split-integrity audits.

The spectral results followed a locked paired post-analysis. The 3,000-row source file was hash-verified before and after analysis. For each dataset, attack model, metric, and predefined variant comparison, observations were paired by protection seed and split seed. Inference used a Wilcoxon signed-rank test on protection-seed cluster means, deterministic 5,000-replicate hierarchical-bootstrap confidence intervals that resampled protection-seed clusters and then split-level observations within clusters, and Holm adjustment globally and within metric. The post-analysis generated 20 summary rows, 48 paired-test rows across Macro-F1, balanced accuracy, and accuracy, and a 16-row

Macro-F1 subset.

The broadest released-view attack applied an explicit `TAFP_RELEASE_VIEW_V1` projection before model training. Node keys were replaced with protected labels or pseudonyms; nodes retained only `entity_type`, and edges retained only the released protected relation type. Raw text, original node and relation labels, sensitivity indicators and scores, and audit metadata were removed. A two-layer GCN graph autoencoder with 64 hidden units, a 32-dimensional output, and 160 training epochs generated node representations. Endpoint-pair classification used a class-balanced (64, 32) multilayer perceptron under three feature modes: topology only; topology plus entity type; and topology plus entity type with target-edge-excluded distributions of neighboring released protected relation labels.

The released-view GCN-GAE protocol evaluated PubMed and SynthPII under the baseline, full TAFP-GraphRAG, and `random_full_budget` conditions across 15 protection seeds, ten locked split seeds, and three feature modes, yielding 2,700 model evaluations. GCN representations were learned transductively from the complete released topology, so target-edge existence remained observable. Relation-classifier training and test edges were disjoint, and the target edge's own protected relation label was excluded from all pair-specific released-attribute features. Macro-F1, balanced accuracy, and accuracy were reported.

5.5. Structural and Retrieval-Grounded Utility Evaluation

Structural retrieval support was evaluated using query definitions fixed on each original graph before protection. Protocol seed 20260618 deterministically ordered candidate nodes. For each dataset, 50 source nodes were selected and up to three targets sampled at each original shortest-path distance of one, two, and three hops. The protocol comprised 2,208 fixed source–target paths and 750 unique source–hop contexts across five datasets. The same paths, sources, targets, and hop radii were reused for all 15 protection seeds and seven primary variants, yielding 231,840 path-level records and 78,750 relevant-subgraph records.

For each fixed path, query-support edge retention was the fraction of original path edges remaining between the same endpoints after transformation. The protocol also recorded complete original-path preservation; endpoint reachability; reachability within the original hop count or within three hops; protected shortest-path length; and path stretch relative to the original hop count. Original-path records further measured retained sensitive- and non-sensitive-edge rates, concealment of original sensitive-relation labels, and retention of original non-sensitive relation semantics.

Relevant-subgraph utility was evaluated separately for every fixed source and hop radius. The original context contained nodes reachable from the source at distances from one through the specified hop radius and the edges induced by those nodes together with the source. The corresponding context was reconstructed on each transformed graph. Comparisons used relevant-node recall, precision, and Jaccard similarity, together with relevant-edge recall and Jaccard similarity. These measures quantify structural support for the fixed retrieval tasks; they do not by themselves establish generated-answer correctness, completeness, or semantic equivalence.

A controlled model-facing relation-token protocol tested whether an instruction-tuned generator propagated the relation label already exposed in transformed graph context. It used 200 fixed endpoint pairs, 40 per dataset, and evaluated baseline, `relation_only`, full, `random_relation_budget`, and `random_full_budget`. The seed-42 run used Qwen/Qwen2.5-3B-Instruct with deterministic decoding (do_sample=False) and at most 12 generated tokens, producing 1,000 outputs. Context contained protected node labels, entity types, node-text fields truncated to 180 characters, the exposed source–target relation, and deterministic local-edge evidence. The prompt required exactly one label from a query-specific allowed-label list. This protocol isolates transmission of an already exposed relation token and is distinct from both the released-view attacker and the locked retrieval-grounded end-to-end task.

The corresponding multi-seed layer evaluated the exact exposed source–target relation for the same 200 endpoint pairs across the same five variants and 15 protection seeds, producing 15,000 transformed-context records without additional language-model generation. The seed-42 records were joined to the 1,000 generated examples by query, dataset, seed, and variant so that the relation state supplied to the generator could be checked against the exact transformed graph.

The locked retrieval-grounded end-to-end protocol used 718 fixed queries selected from the pre-protection path protocol. The manifest assigned 218 queries to non-sensitive utility and 500 to sensitive disclosure; 268 had a one-hop budget, 251 a two-hop budget, and 199 a three-hop budget. Each query was evaluated under baseline, full, `random_full_budget`, and sensitivity-blind 2K conditions for 15 protection seeds, producing 43,080 generations.

For each query–condition–seed tuple, deterministic breadth-first search ran from source to target with the locked hop budget as cutoff. Neighbors were expanded in lexicographic order of their string identifiers, and the first shortest path was returned. When no path was found within the cutoff, the exposed sequence was `NOT_CONNECTED`; otherwise, path edges were serialized from source to target. The completed full protocol also added at most one deterministic side edge per retrieved path node after excluding path edges and previously selected duplicates. Graph-derived prompt context contained protected node labels or pseudonyms, permitted entity types, ordered protected-relation evidence, and transformed connectivity; raw source text, sensitivity fields, and audit metadata were excluded. Qwen/Qwen2.5-3B-Instruct generated relation-label-only outputs using deterministic decoding, a fixed generation seed of 42, a maximum input length of 3,072 tokens, and at most 32 generated tokens.

Non-sensitive utility measures were exact relation-sequence match; multiset relation-token precision, recall, and F1; retrieved-evidence recall; exact faithfulness to the retrieved relation sequence; and faithfulness-token F1. Sensitive disclosure measures were generation of any original sensitive relation token, sensitive-token recall, exposure of original sensitive tokens in retrieved evidence, and response with the protected marker. The execution audit checked the locked query set, expected row count, duplicate records, parse failures, query-set drift, applicable exact-degree invariants, and the 2K joint-degree invariant. Statistical pairing, multiplicity correction, audit layers, and artifact-level reproducibility controls are specified in Section 5.6.

The locked prompt contract instructed the generator to use only the numbered `RETRIEVED ORDERED PATH EVIDENCE` lines, copy each relation label between the two separators in order, join labels with " > ", output exactly `NOT_CONNECTED` when no path was found, and return a single line without node identifiers, explanations, JSON, bullets, or code fences. The parser removed common answer prefixes and code fences, normalized arrow, pipe, and comma delimiters, mapped tokens only to the query-specific allowed relation set, and emitted `PARSE_ERROR` when no allowed relation was recovered. All parse failures remained in the primary analysis.

5.6. Statistical Analysis, Audit Layers, and Reproducibility

Unless a protocol retained an explicit train–test split hierarchy, inferential comparisons were paired at the protection-seed level. Compared conditions were aligned by dataset and locked seed identifier; incomplete seed sets, duplicate condition rows, or missing paired values caused the corresponding audit to fail. Descriptive graph- and utility-level summaries report means and standard deviations across the 15 protection seeds defined for each protocol.

For seed-paired graph, semantic-selectivity, controlled relation-token, global-fidelity, hyperparameter, 2K, marker-leakage, policy-error, and path/subgraph analyses, the principal procedure was a two-sided exact sign-flip test on paired seed differences. Ninety-five-percent percentile bootstrap intervals for the mean paired difference used 20,000 resamples in the controlled relation-token, global-fidelity, hyperparameter, 2K, marker-leakage, and policy-error analyses, and 50,000 in the semantic-selectivity and structural path/subgraph analyses. Where produced by protocol-specific code, two-sided Wilcoxon signed-rank results, Cohen's d_z, and matched rank-biserial effect sizes were retained as complementary diagnostics. Interpretation relied on the paired difference, interval estimate, and multiplicity-adjusted result rather than an unadjusted p-value alone.

Holm correction was applied within pre-specified protocol families rather than across unrelated study outcomes. Path and relevant-subgraph analyses corrected six full-versus-comparator tests within each dataset–metric family. The primary semantic-selectivity table corrected two matched-control comparisons within each dataset–metric family. Controlled relation-token and global-fidelity tables used protocol-wide families of 20 and 60 tests. Hyperparameter tests were corrected within entity-selection, relation-rewiring, and topology-rewiring families of 10, 40, and 40 tests. The 2K analysis used separate semantic-selectivity and structural-fidelity families of 30 and 90 tests. Sensitivity-marker and policy-misclassification comparisons were corrected separately within each reported metric. The two-sided family-wise threshold after Holm adjustment was 0.05.

The spectral-embedding analysis retained its nested split structure. Each comparison contained 150 paired observations from 15 protection seeds and ten split seeds. Differences were averaged across split seeds within each protection seed for a two-sided Wilcoxon signed-rank test over 15 protection-seed clusters. The 95% hierarchical-bootstrap interval used 5,000 replicates, resampling protection-seed clusters and then split observations within each sampled cluster. Holm-adjusted values were reported across the complete 48-test family and within each metric. For the released-view GCN-GAE analysis, ten split-level outcomes were averaged within each protection seed before exact sign-flip and Wilcoxon tests were applied to the 15 paired seed means; Holm correction covered the 36 reported comparisons.

For end-to-end relation-question answering, query-level observations were first averaged within dataset, protection seed, variant, and task. Overall summaries then averaged dataset-level values within each protection seed. Means, standard deviations, and 95% t-intervals were calculated across the 15 seed means. Full TAFP-GraphRAG was compared with baseline, `random_full_budget`, and 2K using two-sided Wilcoxon signed-rank tests over matched seed means. Holm adjustment was applied within each scope–task–metric family across the three comparators. The primary analysis retained all 43,080 generated records, including 26 parse failures (0.06035%); a parse-success-only analysis served as a sensitivity check.

Runtime was summarized by dataset and variant as the mean and standard deviation across protection seeds, both in seconds and normalized per 1,000 edges. Audit evidence was maintained at three distinct levels. First, the implementation-level `protection_audit` stored the numerical transformation summaries defined in Section 4. Second, run-level result and audit tables checked expected and observed row counts, duplicate and missing records, candidate and class-balance integrity, fixed query sets, determinism, degree-preservation constraints, and the 2K joint-degree invariant. Third, artifact-level manifests linked source files, result tables, supplementary materials, and analysis outputs through SHA-256 identities.

No standalone row-level `protection_audit` CSV was generated; its absence is explicitly recorded and is not treated as a failed protocol audit. Fixed-edge, multisplit, structural-utility, and end-to-end execution audits were evaluated separately and passed their stated integrity checks. The archived environment record identifies Python 3.12.13, NetworkX 3.6.1, NumPy 2.4.4, pandas 3.0.2, scikit-learn 1.8.0, SciPy 1.17.1, and PyTorch 2.11.0. Exact source identities, dependency records, table-generation manifests, analysis manifests, supplementary-package manifests, and submission-package manifests were retained with cryptographic hashes.

6. Results

6.1. Selective Concealment of Sensitive Relation Semantics

Across the five evaluated datasets, full TAFP-GraphRAG concealed 0.99993–1.00000 of the original labels of policy-designated sensitive relations while retaining 0.89935–0.92193 of non-sensitive relation semantics (Table 2). Exact per-node degree preservation remained 1.00000 in every full-protection condition, with original-edge retention of 0.81833–0.86109. Relation-only achieved complete sensitive-label concealment and complete non-sensitive semantic retention on every dataset, with original-edge retention of 0.88643–0.93502. Thus, non-sensitive retention fell from perfect to approximately 0.90–0.92 only after topology perturbation was added to the semantic protection layer.

The perturbation-budget-matched random control preserved nearly the same fraction of original edges as full TAFP-GraphRAG: the largest absolute full-minus-random difference in original-edge retention was 0.00044, and none of the five paired differences remained significant after Holm correction. Despite the matched structural budget, random full-budget perturbation concealed only 0.57838–0.96659 of sensitive labels and retained 0.03355–0.41804 of non-sensitive semantics. Full TAFP-GraphRAG was superior in all ten dataset-by-semantic-metric comparisons, with a Holm-adjusted p value of 0.000122 for each comparison (Supplementary Table S10). This separation shows that the semantic advantage was attributable to sensitivity-guided selection rather than to perturbation volume or original-edge retention alone.

The minority-sensitive ablation produced the same qualitative protection profile at 5%, 10%, 20%, and 30% policy-designated sensitive-edge prevalence (Supplementary Table S27). Across these settings, full TAFP-GraphRAG maintained mean sensitive-label concealment of 0.9995–0.9998, mean non-sensitive semantic retention of 0.9200–0.9206, original-edge retention of 0.8876–0.9149, and exact degree preservation of 1.0000. At 5% prevalence, the random relation-budget control retained more non-sensitive semantics than the full method (0.9499 versus 0.9202) but concealed only 0.0475 of sensitive labels, compared with 0.9996 for full TAFP-GraphRAG. From 10% through 30% prevalence, the full method achieved both higher concealment and higher non-sensitive retention than that control; the random full-budget control remained lower on both semantic metrics at every prevalence level. The ablation therefore supports a prevalence-stable concealment–retention profile, while the advantage over sensitivity-blind controls remains prevalence dependent.

Figure 4 summarizes the graph-level semantic-selectivity and utility comparison, and Figure 5 summarizes the minority-sensitive policy ablation.

Table 2: Protection integrity and semantic selectivity across datasets and variants. Values are means ± standard deviations over 15 protection seeds. Exact degree preservation equals one only when every node retained its original degree.

Dataset	Variant	Original sensitive-label concealment	Non-sensitive semantic retention	Original-edge retention	Exact per-node degree preservation
AGNews	baseline	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
AGNews	full	1.0000 ± 0.0001	0.9196 ± 0.0026	0.8611 ± 0.0010	1.0000 ± 0.0000
AGNews	`random_edge_dropout`	0.0787 ± 0.0042	0.9189 ± 0.0050	0.9202 ± 0.0000	0.0000 ± 0.0000
AGNews	`random_full_budget`	0.5784 ± 0.0059	0.4180 ± 0.0069	0.8607 ± 0.0006	1.0000 ± 0.0000
AGNews	`random_relation_budget`	0.5413 ± 0.0056	0.4551 ± 0.0066	0.9350 ± 0.0001	1.0000 ± 0.0000
AGNews	`relation_only`	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9350 ± 0.0001	1.0000 ± 0.0000
AGNews	`topology_only`	0.0804 ± 0.0022	0.9211 ± 0.0026	0.9204 ± 0.0002	1.0000 ± 0.0000
CNN_DM	baseline	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN_DM	full	1.0000 ± 0.0000	0.9171 ± 0.0093	0.8188 ± 0.0010	1.0000 ± 0.0000
CNN_DM	`random_edge_dropout`	0.0802 ± 0.0005	0.9221 ± 0.0051	0.9200 ± 0.0000	0.0000 ± 0.0000
CNN_DM	`random_full_budget`	0.9243 ± 0.0008	0.0730 ± 0.0094	0.8192 ± 0.0006	1.0000 ± 0.0000
CNN_DM	`random_relation_budget`	0.9177 ± 0.0010	0.0798 ± 0.0108	0.8900 ± 0.0001	1.0000 ± 0.0000
CNN_DM	`relation_only`	1.0000 ± 0.0000	1.0000 ± 0.0000	0.8900 ± 0.0001	1.0000 ± 0.0000
CNN_DM	`topology_only`	0.0798 ± 0.0008	0.9179 ± 0.0085	0.9201 ± 0.0001	1.0000 ± 0.0000
IMDB	baseline	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
IMDB	full	0.9999 ± 0.0001	0.9192 ± 0.0047	0.8429 ± 0.0012	1.0000 ± 0.0000
IMDB	`random_edge_dropout`	0.0791 ± 0.0019	0.9180 ± 0.0044	0.9200 ± 0.0000	0.0000 ± 0.0000
IMDB	`random_full_budget`	0.7251 ± 0.0033	0.2754 ± 0.0075	0.8429 ± 0.0009	1.0000 ± 0.0000
IMDB	`random_relation_budget`	0.7017 ± 0.0030	0.3004 ± 0.0071	0.9159 ± 0.0001	1.0000 ± 0.0000
IMDB	`relation_only`	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9160 ± 0.0002	1.0000 ± 0.0000
IMDB	`topology_only`	0.0804 ± 0.0020	0.9211 ± 0.0046	0.9201 ± 0.0001	1.0000 ± 0.0000
PubMed	baseline	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
PubMed	full	1.0000 ± 0.0000	0.8993 ± 0.0222	0.8186 ± 0.0013	1.0000 ± 0.0000
PubMed	`random_edge_dropout`	0.0795 ± 0.0008	0.9081 ± 0.0224	0.9201 ± 0.0000	0.0000 ± 0.0000
PubMed	`random_full_budget`	0.9666 ± 0.0011	0.0336 ± 0.0178	0.8189 ± 0.0015	1.0000 ± 0.0000
PubMed	`random_relation_budget`	0.9640 ± 0.0007	0.0357 ± 0.0186	0.8858 ± 0.0006	1.0000 ± 0.0000

Dataset	Variant	Original sensitive-label concealment	Non-sensitive semantic retention	Original-edge retention	Exact per-node degree preservation
PubMed	`relation_only`	1.0000 ± 0.0000	1.0000 ± 0.0000	0.8864 ± 0.0005	1.0000 ± 0.0000
PubMed	`topology_only`	0.0788 ± 0.0012	0.9020 ± 0.0320	0.9209 ± 0.0003	1.0000 ± 0.0000
SynthPII	baseline	0.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
SynthPII	full	1.0000 ± 0.0000	0.9219 ± 0.0132	0.8183 ± 0.0012	1.0000 ± 0.0000
SynthPII	`random_edge_dropout`	0.0801 ± 0.0016	0.9209 ± 0.0178	0.9200 ± 0.0000	0.0000 ± 0.0000
SynthPII	`random_full_budget`	0.9256 ± 0.0019	0.0722 ± 0.0210	0.8188 ± 0.0016	1.0000 ± 0.0000
SynthPII	`random_relation_budget`	0.9194 ± 0.0017	0.0784 ± 0.0199	0.8901 ± 0.0001	1.0000 ± 0.0000
SynthPII	`relation_only`	1.0000 ± 0.0000	1.0000 ± 0.0000	0.8901 ± 0.0002	1.0000 ± 0.0000
SynthPII	`topology_only`	0.0801 ± 0.0017	0.9209 ± 0.0189	0.9201 ± 0.0001	1.0000 ± 0.0000

6.2. Exact Degree Preservation and Structural Effects

Across the 450 dataset–seed–variant rows for the baseline and the five degree-preserving perturbation variants, every row had zero changed-degree nodes, zero degree-sequence L1 error, and exact per-node degree preservation equal to 1. By contrast, none of the 75 matched random edge-dropout rows preserved every node degree; changed-degree nodes ranged from 295 to 1,690 and degree-sequence L1 error from 384 to 2,440 (Supplementary Table S6). Edge-retention volume did not explain this difference. Topology-only retained 0.92005–0.92089 of original edges, whereas random edge dropout retained 0.91999–0.92021; only the former preserved every node degree (Table 2). Full and random full-budget also preserved every node degree while retaining 0.81833–0.86109 and 0.81878–0.86071 of original edges, respectively.

The 2K full-budget control imposed a stronger structural constraint: across five datasets and 15 seeds per dataset, its exact degree-sequence rate was 1.0 and mean joint-degree L1 error was 0 for every dataset (Supplementary Table S17). The locked summaries also reported zero component-count deviation and zero largest-component-ratio deviation. These constraints did not imply exact preservation of all higher-order structure: mean absolute clustering-coefficient deviation ranged from 0.00089 to 0.28649 and mean absolute transitivity deviation from 0.00082 to 0.02185. Full and random full-budget preserved individual node degrees but not the joint-degree distribution; their dataset-level mean joint-degree L1 errors ranged from 174.4 to 1,692.9 and from 160.0 to 1,695.6, respectively.

Despite the stronger 2K invariant, full TAFP-GraphRAG was better in all ten paired comparisons of sensitive-relation concealment and non-sensitive semantic retention, with Holm-adjusted $p = 0.001831$ for each comparison (Supplementary Table S18). None of the ten corresponding 2K-versus-random-full-budget comparisons was significant after Holm correction. Thus, exact individual- and joint-degree preservation constrained structural change but did not reproduce sensitivity-selective semantic protection.

Table 3 reports the transformation-runtime and edit-integrity summaries.

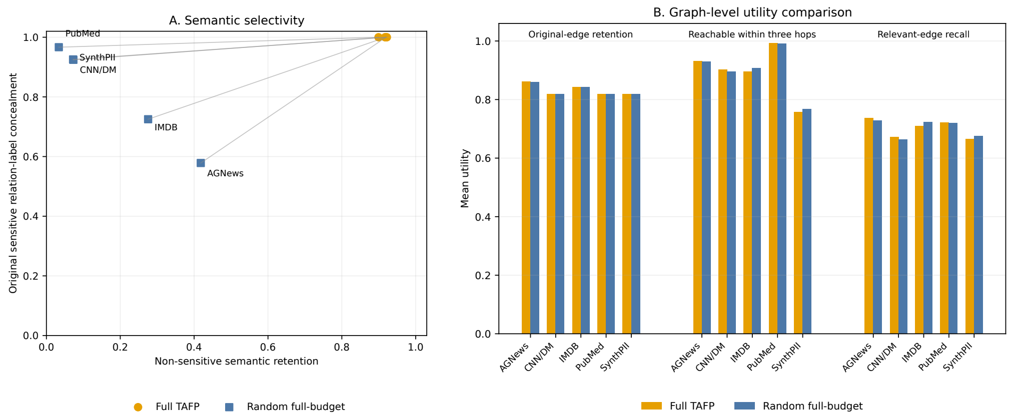

Figure 4: Graph-level semantic selectivity and utility across the five evaluated datasets. Panel A plots mean non-sensitive semantic retention against mean original sensitive relation-label concealment for Full TAFP and the perturbation-budget-matched random full-budget control. Lines connect the two variants within each dataset. Panel B reports three manuscript-defined graph-level utility measures—original-edge retention, reachability within three hops, and relevant-edge recall—using the same two variants. Values are means over the locked protection-seed protocol; dispersion statistics remain available in the corresponding tables.

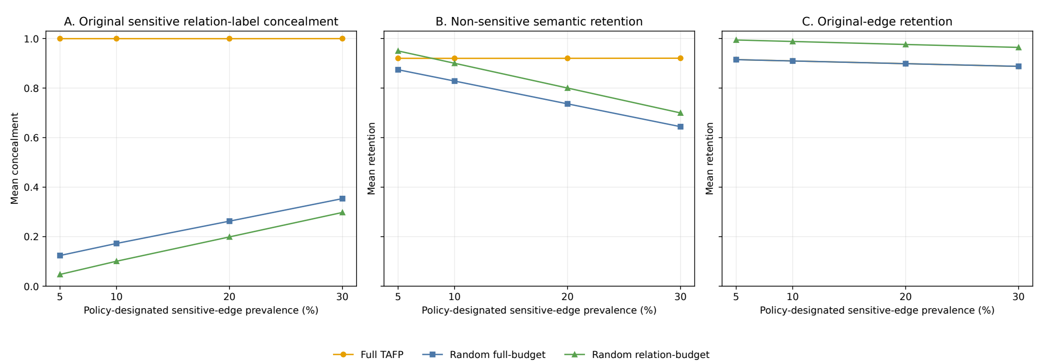

Figure 5: Minority-sensitive policy ablation across 5%, 10%, 20%, and 30% policy-designated sensitive-edge prevalence. Panel A reports original sensitive relation-label concealment, Panel B reports non-sensitive semantic retention, and Panel C reports original-edge retention for Full TAFP and the two sensitivity-blind random controls. Points are unweighted means across the five evaluated datasets, and whiskers show the minimum-to-maximum range of the dataset-level means. Each dataset-level value summarizes the locked 15-seed protection protocol. Exact per-node degree preservation was 1.0000 for every displayed prevalence, dataset, and variant configuration.

Table 3: Transformation runtime and edit integrity. Runtime covers apply_variant() only; values are means ± standard deviations over 15 protection seeds.

Dataset	Variant	Runtime (s)	Runtime/1000 edges (s)	Added edges	Removed original edges	Changed-degree nodes
AGNews	baseline	0.0580 ± 0.0348	0.0089 ± 0.0053	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
AGNews	full	0.1354 ± 0.0452	0.0207 ± 0.0069	908.7333 ± 6.8187	908.7333 ± 6.8187	0.0000 ± 0.0000
AGNews	`random_edge_dropout`	0.0911 ± 0.0560	0.0139 ± 0.0086	0.0000 ± 0.0000	522 ± 0.0000	745.8000 ± 9.7848
AGNews	`random_full_budget`	0.1218 ± 0.0345	0.0186 ± 0.0053	911.2667 ± 3.6541	911.2667 ± 3.6541	0.0000 ± 0.0000
AGNews	`random_relation_budget`	0.0851 ± 0.0312	0.0130 ± 0.0048	425.4667 ± 0.8338	425.4667 ± 0.8338	0.0000 ± 0.0000
AGNews	`relation_only`	0.0968 ± 0.0438	0.0148 ± 0.0067	425.0667 ± 0.8837	425.0667 ± 0.8837	0.0000 ± 0.0000
AGNews	`topology_only`	0.0784 ± 0.0364	0.0120 ± 0.0056	520.6000 ± 1.1832	520.6000 ± 1.1832	0.0000 ± 0.0000
CNN_DM	baseline	0.1132 ± 0.0428	0.0074 ± 0.0028	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
CNN_DM	full	0.3300 ± 0.0294	0.0216 ± 0.0019	2762.3333 ± 14.4897	2762.3333 ± 14.4897	0.0000 ± 0.0000
CNN_DM	`random_edge_dropout`	0.1507 ± 0.0529	0.0099 ± 0.0035	0.0000 ± 0.0000	1,220 ± 0.0000	1663.8000 ± 16.2226
CNN_DM	`random_full_budget`	0.3363 ± 0.0397	0.0221 ± 0.0026	2756.8000 ± 9.5559	2756.8000 ± 9.5559	0.0000 ± 0.0000
CNN_DM	`random_relation_budget`	0.2315 ± 0.0594	0.0152 ± 0.0039	1677.9333 ± 1.2799	1677.9333 ± 1.2799	0.0000 ± 0.0000
CNN_DM	`relation_only`	0.2150 ± 0.0567	0.0141 ± 0.0037	1677.9333 ± 1.4376	1677.9333 ± 1.4376	0.0000 ± 0.0000
CNN_DM	`topology_only`	0.1802 ± 0.0545	0.0118 ± 0.0036	1219.0667 ± 0.9612	1219.0667 ± 0.9612	0.0000 ± 0.0000
IMDB	baseline	0.0585 ± 0.0196	0.0072 ± 0.0024	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
IMDB	full	0.1848 ± 0.0478	0.0228 ± 0.0059	1,273 ± 9.4868	1,273 ± 9.4868	0.0000 ± 0.0000
IMDB	`random_edge_dropout`	0.0831 ± 0.0389	0.0103 ± 0.0048	0.0000 ± 0.0000	648 ± 0.0000	908.5333 ± 14.5498

Dataset	Variant	Runtime (s)	Runtime/1000 edges (s)	Added edges	Removed original edges	Changed-degree nodes
IMDB	`random_full_budget`	0.1993 ± 0.0543	0.0246 ± 0.0067	1272.7333 ± 7.4014	1272.7333 ± 7.4014	0.0000 ± 0.0000
IMDB	`random_relation_budget`	0.1482 ± 0.0594	0.0183 ± 0.0073	681.3333 ± 0.8997	681.3333 ± 0.8997	0.0000 ± 0.0000
IMDB	`relation_only`	0.1149 ± 0.0438	0.0142 ± 0.0054	680.4667 ± 1.2459	680.4667 ± 1.2459	0.0000 ± 0.0000
IMDB	`topology_only`	0.1153 ± 0.0508	0.0142 ± 0.0063	647.6667 ± 0.6172	647.6667 ± 0.6172	0.0000 ± 0.0000
PubMed	baseline	0.0235 ± 0.0044	0.0055 ± 0.0010	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PubMed	full	0.0797 ± 0.0417	0.0187 ± 0.0098	771.6667 ± 5.6904	771.6667 ± 5.6904	0.0000 ± 0.0000
PubMed	`random_edge_dropout`	0.0408 ± 0.0297	0.0096 ± 0.0070	0.0000 ± 0.0000	340 ± 0.0000	343 ± 10.1489
PubMed	`random_full_budget`	0.0842 ± 0.0384	0.0198 ± 0.0090	770.3333 ± 6.5320	770.3333 ± 6.5320	0.0000 ± 0.0000
PubMed	`random_relation_budget`	0.0544 ± 0.0222	0.0128 ± 0.0052	485.8667 ± 2.5598	485.8667 ± 2.5598	0.0000 ± 0.0000
PubMed	`relation_only`	0.0512 ± 0.0300	0.0120 ± 0.0071	483.1333 ± 2.2636	483.1333 ± 2.2636	0.0000 ± 0.0000
PubMed	`topology_only`	0.0422 ± 0.0267	0.0099 ± 0.0063	336.5333 ± 1.4075	336.5333 ± 1.4075	0.0000 ± 0.0000
SynthPII	baseline	0.0148 ± 0.0033	0.0062 ± 0.0014	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
SynthPII	full	0.0524 ± 0.0319	0.0218 ± 0.0133	436 ± 2.9761	436 ± 2.9761	0.0000 ± 0.0000
SynthPII	`random_edge_dropout`	0.0180 ± 0.0035	0.0075 ± 0.0015	0.0000 ± 0.0000	192 ± 0.0000	303.7333 ± 5.8975
SynthPII	`random_full_budget`	0.0355 ± 0.0079	0.0148 ± 0.0033	434.9333 ± 3.7506	434.9333 ± 3.7506	0.0000 ± 0.0000
SynthPII	`random_relation_budget`	0.0300 ± 0.0050	0.0125 ± 0.0021	263.8667 ± 0.3519	263.8667 ± 0.3519	0.0000 ± 0.0000
SynthPII	`relation_only`	0.0257 ± 0.0053	0.0107 ± 0.0022	263.7333 ± 0.5936	263.7333 ± 0.5936	0.0000 ± 0.0000
SynthPII	`topology_only`	0.0270 ± 0.0286	0.0113 ± 0.0119	191.8667 ± 0.3519	191.8667 ± 0.3519	0.0000 ± 0.0000

6.3. Attack-Specific Privacy Evaluation

The fixed-edge analysis separated concealment of the original sensitive relation label from inference of original edge existence. Under both relation-only and full, mean recovery of the original sensitive relation type was 0 on all five datasets (Table 4), but structural evidence of prior connectivity remained. For full, local structural AUC ranged from 0.49909 to 0.86879 and hybrid edge-inference AUC from 0.90705 to 0.97351. Thus, direct sensitive-label recovery was removed in this protocol, whereas residual edge-existence inference remained dataset dependent and substantial for several datasets.

The structure-only relation attack was similarly conditional. On PubMed, full TAFP-GraphRAG reduced mean Macro-F1 from 0.64602 for the unprotected baseline and 0.62377 for random edge dropout to 0.61108; the one-sided Holm-adjusted p values were 0.005859 and 0.003906, respectively (Table 5). Full was not significantly different from either perturbation-budget-matched random control. On SynthPII, full achieved Macro-F1 of 0.08635, and none of the four reported full-versus-comparator tests was significant after Holm correction. Topology-only relation inference was therefore weaker on PubMed than under unprotected and deletion-only references, without establishing a sensitivity-guided advantage over matched random perturbation.

Embedding-based attacks reinforced this dataset dependence. On PubMed, spectral-logistic Macro-F1 decreased from 0.72164 under the baseline to 0.57853 under full protection, and spectral-MLP Macro-F1 from 0.69662 to 0.57612. Both full-versus-baseline comparisons remained significant after global Holm correction, whereas neither full-versus-random-full-budget comparison was significant (Supplementary Tables S22–S23). On SynthPII, no consistent reduction appeared: spectral-logistic Macro-F1 was slightly higher under full protection than under the baseline (0.11364 versus 0.10118; globally Holm-adjusted $p = 0.015808$), while the spectral-MLP comparison was not significant. The spectral results therefore support reduced PubMed attack performance, but not a general advantage across datasets or over the matched random control.

The released-view GCN-GAE stress test produced the same boundary. On PubMed, full protection reduced Macro-F1 from 0.76515 to 0.59832 with topology-only features, from 0.88081 to 0.80088 after adding entity type, and from 0.99351 to 0.85222 with the complete permitted released-attribute feature set. All nine PubMed full-versus-baseline metric comparisons were significant after exact sign-flip Holm correction; none of the nine corresponding full-versus-random-full-budget comparisons was significant. On SynthPII, none of the full-versus-baseline or full-versus-random-full-budget comparisons remained significant after correction (Supplementary Tables S28–S29). These learned-representation attacks therefore showed reduced inference relative to the unprotected PubMed graph, but not an effect uniquely attributable to sensitivity-guided selection under the matched structural budget.

Table 6 reports the released-view GCN-GAE results, and Figure 6 summarizes the attack-specific evidence across the evaluated attacker families.

The marker audit identified a separate release-policy risk. Both the default PRIVATE_RELATION marker and relation-label omission yielded policy-membership AUC of 1.0 on every dataset. Applying the same coarse marker to 50% of non-sensitive relations reduced AUC to 0.74597–0.75561, but reduced non-sensitive semantic retention to 0.44401–0.47496. Applying the cover to all non-sensitive relations reduced AUC to 0.5 and non-sensitive semantic retention to 0. Across all cover settings, sensitive-label concealment and exact degree preservation remained 1.0. All 50 paired AUC or retention comparisons against the default marker were significant after Holm correction ($p = 0.001526$). Marker privacy is therefore not obtained without a direct semantic-utility cost under the evaluated release policies.

Finally, the policy-misclassification experiment distinguished correct execution of the supplied policy from protection under the original oracle policy. Supplied-policy marker AUC and supplied-policy relation concealment remained 1.0 in every scenario, and exact edge-count and degree preservation also remained 1.0. At a 20% false-negative rate, oracle sensitive-relation concealment fell to 0.79979–0.80280. At a 20% false-positive rate, oracle non-sensitive label preservation fell to 0.78791–0.80362. Under 20% symmetric error, oracle sensitive concealment was 0.79742–0.80184 and oracle non-sensitive preservation was 0.79216–0.80656 (Supplementary Table S21). TAFP-GraphRAG therefore executed the supplied policy faithfully but did not correct errors in that policy, making policy quality an explicit operational dependency.

Table 4: Fixed-edge evaluation on sensitive target edges. Local structural AUC and hybrid edge AUC represent distinct attacker feature sets. Values are means ± standard deviations over 15 protection seeds.

Dataset	Variant	Local structural AUC	Hybrid edge AUC	Positive-edge retention	Original relation-label recovery	Conditional relation recovery
AGNews	baseline	0.9778 ± 0.0149	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000
AGNews	full	0.8688 ± 0.0211	0.9735 ± 0.0095	0.8087 ± 0.0302	0.0000 ± 0.0000	0.0027 ± 0.0046
AGNews	`random_edge_dropout`	0.9329 ± 0.0236	0.9938 ± 0.0052	0.9193 ± 0.0306	0.9193 ± 0.0306	0.0000 ± 0.0000
AGNews	`random_full_budget`	0.8745 ± 0.0249	0.9846 ± 0.0083	0.8873 ± 0.0371	0.4367 ± 0.0429	0.0060 ± 0.0074
AGNews	`random_relation_budget`	0.9284 ± 0.0219	0.9946 ± 0.0050	0.9480 ± 0.0293	0.4653 ± 0.0416	0.0027 ± 0.0046
AGNews	`relation_only`	0.9211 ± 0.0226	0.9905 ± 0.0060	0.8720 ± 0.0283	0.0000 ± 0.0000	0.0020 ± 0.0041
AGNews	`topology_only`	0.9321 ± 0.0151	0.9926 ± 0.0070	0.9187 ± 0.0229	0.9187 ± 0.0229	0.0033 ± 0.0049
CNN__DM	baseline	0.9591 ± 0.0132	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000
CNN__DM	full	0.8535 ± 0.0270	0.9731 ± 0.0102	0.8120 ± 0.0426	0.0000 ± 0.0000	0.0020 ± 0.0041
CNN__DM	`random_edge_dropout`	0.9120 ± 0.0183	0.9941 ± 0.0038	0.9207 ± 0.0281	0.9207 ± 0.0281	0.0000 ± 0.0000
CNN__DM	`random_full_budget`	0.8475 ± 0.0241	0.9713 ± 0.0108	0.8213 ± 0.0302	0.0867 ± 0.0261	0.0060 ± 0.0083
CNN__DM	`random_relation_budget`	0.8935 ± 0.0212	0.9887 ± 0.0066	0.8913 ± 0.0264	0.0940 ± 0.0307	0.0033 ± 0.0062
CNN__DM	`relation_only`	0.8957 ± 0.0183	0.9889 ± 0.0046	0.8880 ± 0.0390	0.0000 ± 0.0000	0.0020 ± 0.0041
CNN__DM	`topology_only`	0.9200 ± 0.0143	0.9929 ± 0.0071	0.9313 ± 0.0267	0.9313 ± 0.0267	0.0007 ± 0.0026
IMDB	baseline	0.8647 ± 0.0262	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000
IMDB	full	0.8135 ± 0.0288	0.9664 ± 0.0138	0.8207 ± 0.0457	0.0000 ± 0.0000	0.0087 ± 0.0092
IMDB	`random_edge_dropout`	0.8282 ± 0.0254	0.9831 ± 0.0094	0.9147 ± 0.0226	0.9147 ± 0.0226	0.0000 ± 0.0000
IMDB	`random_full_budget`	0.8338 ± 0.0305	0.9734 ± 0.0130	0.8427 ± 0.0359	0.2893 ± 0.0287	0.0087 ± 0.0083
IMDB	`random_relation_budget`	0.8652 ± 0.0346	0.9881 ± 0.0075	0.9100 ± 0.0285	0.3120 ± 0.0303	0.0067 ± 0.0082
IMDB	`relation_only`	0.8466 ± 0.0225	0.9800 ± 0.0105	0.8820 ± 0.0314	0.0000 ± 0.0000	0.0080 ± 0.0068
IMDB	`topology_only`	0.8609 ± 0.0310	0.9904 ± 0.0071	0.9173 ± 0.0234	0.9173 ± 0.0234	0.0047 ± 0.0052
PubMed	baseline	0.8891 ± 0.0270	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000
PubMed	full	0.7194 ± 0.0434	0.9197 ± 0.0334	0.8087 ± 0.0461	0.0000 ± 0.0000	0.0607 ± 0.0252
PubMed	`random_edge_dropout`	0.8410 ± 0.0285	0.9895 ± 0.0067	0.9173 ± 0.0317	0.9173 ± 0.0317	0.0000 ± 0.0000
PubMed	`random_full_budget`	0.7244 ± 0.0295	0.9268 ± 0.0141	0.8353 ± 0.0210	0.0407 ± 0.0167	0.0667 ± 0.0176
PubMed	`random_relation_budget`	0.7834 ± 0.0367	0.9570 ± 0.0104	0.8913 ± 0.0245	0.0433 ± 0.0188	0.0453 ± 0.0136
PubMed	`relation_only`	0.7800 ± 0.0389	0.9571 ± 0.0206	0.8827 ± 0.0358	0.0000 ± 0.0000	0.0433 ± 0.0195

Dataset	Variant	Local structural AUC	Hybrid edge AUC	Positive-edge retention	Original relation-label recovery	Conditional relation recovery
PubMed	`topology_only`	0.8119 ± 0.0292	0.9701 ± 0.0117	0.9200 ± 0.0338	0.9200 ± 0.0338	0.0367 ± 0.0154
SynthPII	baseline	0.4897 ± 0.0129	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.0000 ± 0.0000
SynthPII	full	0.4991 ± 0.0108	0.9070 ± 0.0189	0.8113 ± 0.0338	0.0000 ± 0.0000	0.0007 ± 0.0026
SynthPII	`random_edge_dropout`	0.4916 ± 0.0113	0.9594 ± 0.0127	0.9187 ± 0.0245	0.9187 ± 0.0245	0.0000 ± 0.0000
SynthPII	`random_full_budget`	0.4945 ± 0.0139	0.9085 ± 0.0139	0.8180 ± 0.0243	0.0580 ± 0.0197	0.0020 ± 0.0056
SynthPII	`random_relation_budget`	0.4921 ± 0.0208	0.9493 ± 0.0133	0.8993 ± 0.0258	0.0640 ± 0.0203	0.0020 ± 0.0056
SynthPII	`relation_only`	0.4964 ± 0.0111	0.9355 ± 0.0118	0.8753 ± 0.0223	0.0000 ± 0.0000	0.0013 ± 0.0035
SynthPII	`topology_only`	0.4941 ± 0.0115	0.9549 ± 0.0139	0.9087 ± 0.0270	0.9080 ± 0.0260	0.0007 ± 0.0026

Table 5: Hierarchical paired comparisons for structure-only relation inference. The Comparator column identifies the reference condition; differences are calculated as full minus comparator. Confidence intervals preserve the split-seed and protection-seed hierarchy, and p-values are Holm adjusted.

Dataset	Metric	Comparator	Full value	Reference value	Difference	95% CI	Holm p	Sig. after Holm
PubMed	Macro-F1	Baseline	0.6111	0.6460	-0.0349	[-0.0483, -0.0205]	0.0059	Yes
PubMed	Macro-F1	Random edge dropout	0.6111	0.6238	-0.0127	[-0.0232, -0.0030]	0.0039	Yes
PubMed	Macro-F1	Random full-budget	0.6111	0.6056	0.0054	[-0.0019, 0.0129]	0.9873	No
PubMed	Macro-F1	Random relation-budget	0.6111	0.6150	-0.0039	[-0.0134, 0.0058]	0.2559	No
PubMed	Balanced accuracy	Baseline	0.6207	0.6510	-0.0304	[-0.0428, -0.0168]	0.0117	Yes
PubMed	Balanced accuracy	Random edge dropout	0.6207	0.6329	-0.0122	[-0.0219, -0.0030]	0.0039	Yes
PubMed	Balanced accuracy	Random full-budget	0.6207	0.6145	0.0062	[-0.0020, 0.0144]	0.9775	No
PubMed	Balanced accuracy	Random relation-budget	0.6207	0.6227	-0.0021	[-0.0113, 0.0073]	0.4629	No
SynthPII	Macro-F1	Baseline	0.0863	0.0849	0.0014	[-0.0120, 0.0139]	1.0000	No
SynthPII	Macro-F1	Random edge dropout	0.0863	0.0878	-0.0015	[-0.0073, 0.0045]	0.8594	No
SynthPII	Macro-F1	Random full-budget	0.0863	0.0831	0.0033	[-0.0026, 0.0090]	1.0000	No
SynthPII	Macro-F1	Random relation-budget	0.0863	0.0805	0.0059	[-0.0005, 0.0120]	1.0000	No
SynthPII	Balanced accuracy	Baseline	0.1023	0.1009	0.0014	[-0.0140, 0.0154]	1.0000	No
SynthPII	Balanced accuracy	Random edge dropout	0.1023	0.1043	-0.0020	[-0.0076, 0.0038]	0.4766	No
SynthPII	Balanced accuracy	Random full-budget	0.1023	0.1018	0.0005	[-0.0056, 0.0069]	1.0000	No
SynthPII	Balanced accuracy	Random relation-budget	0.1023	0.0983	0.0040	[-0.0029, 0.0107]	1.0000	No

Table 6: Released-view GCN-GAE relation-inference performance. Values are means over 15 protection seeds and ten locked split seeds (150 evaluations per cell). The released-attribute mode uses entity type and target-edge-excluded distributions of neighboring exposed protected relation labels.

Dataset	Variant	Topology-only Macro-F1	+ Entity type Macro-F1	+ Released attrs Macro-F1	+ Released attrs Bal. acc.	+ Released attrs Accuracy
PubMed	Baseline	0.7652	0.8808	0.9935	0.9933	0.9951
PubMed	Full TAFP	0.5983	0.8009	0.8522	0.8492	0.8807
PubMed	Random full budget	0.6038	0.7952	0.8335	0.8315	0.8739
SynthPII	Baseline	0.1075	0.1037	0.1091	0.1122	0.1186
SynthPII	Full TAFP	0.1029	0.1094	0.1060	0.1078	0.1137
SynthPII	Random full budget	0.1025	0.1104	0.1078	0.1112	0.1181

6.4. Structural and Retrieval-Grounded Utility

Across the five datasets, full TAFP-GraphRAG retained substantial edge support for the fixed retrieval paths. Mean query-support edge retention ranged from 0.81359 to 0.86568, and the complete original path was preserved for 0.67237–0.75662 of queries (Table 7). Endpoint reachability remained 1.0 in every dataset. Reachability within the original hop budget ranged from 0.71111 to 0.86697 and within three hops from 0.75796 to 0.99246. PubMed showed the strongest three-hop preservation, whereas SynthPII showed the largest structural displacement under the same protocol.

The relevant-subgraph results likewise distinguished retrieval access from exact local-structure preservation. Under full protection, relevant-node recall ranged from 0.72295 to 0.86591 and relevant-edge recall from 0.66491 to 0.73790. The corresponding Jaccard ranges were 0.59406–0.78626 for nodes and 0.50825–0.61775 for edges (Table 8). Most originally relevant nodes and a majority of originally relevant edges therefore remained available, although the transformed contexts were not identical to the original induced subgraphs.

The matched `random_full_budget` control produced broadly similar structural utility. Sixteen of the 20 paired path comparisons and 19 of the 20 paired subgraph comparisons were non-significant after Holm correction (Supplementary Tables S8–S9). The four significant path differences occurred on IMDB and favored the matched random control. The only significant subgraph difference, SynthPII relevant-node recall, was also lower under full protection. These results do not establish a general structural-utility advantage for sensitivity-guided selection over the matched random full-budget condition.

The deletion-only control retained more original structure on most measures. Compared with random edge dropout, full protection had lower means in 19 of 20 path comparisons and all 20 subgraph comparisons; all were significant after Holm correction. This contrast should be interpreted by objective rather than structural overlap alone: random edge dropout directly optimizes neither selective concealment of policy-designated relation semantics nor the degree-preserving behavior evaluated for the designated TAFP variants.

Overall, the fixed-query evaluation shows that TAFP-GraphRAG preserved endpoint access and substantial path and local-subgraph support while reducing exact structural overlap. These metrics quantify retrieval-grounded structural utility only; generated-answer utility and sensitive-relation disclosure are evaluated separately in the end-to-end relation-QA protocol.

Table 7 reports endpoint reachability and reachability within the original and three-hop budgets for the fixed source–target pairs; Supplementary Table S17 reports component-count and largest-component-ratio deviations for the matched 2K analysis. The protocol did not compute a global average shortest-path length, so no whole-graph path-length claim is made.

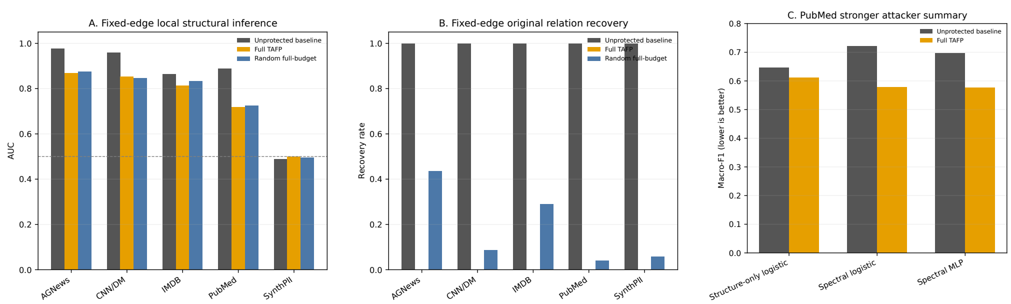

Figure 6: Attack-specific relation-inference evidence. Panel A reports fixed-edge local structural AUC across the five evaluated datasets; the dashed line marks chance-level discrimination. Panel B reports direct recovery of the original relation label after deterministic sanitization for sensitive target edges. Panel C reports the PubMed stronger-attacker summary using only directly reported values: the structure-only logistic attack from Table 5 and the spectral logistic and spectral MLP attackers from Section 6.3. For Panel C, lower Macro-F1 indicates lower relation-inference success. Random full-budget is not shown for the spectral attackers because those exact values are not directly tabulated in the manuscript.

Table 7: Fixed-query path-level structural utility. Endpoint reachability applies to the predefined source-target pairs and is not a global graph-connectivity measure.

Dataset	Variant	Query-support edge retention	Exact path preservation	Endpoint reachability	Reachable within original hop	Reachable within three hops
AGNews	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
AGNews	full	0.8657 ± 0.0244	0.7566 ± 0.0420	1.0000 ± 0.0000	0.8268 ± 0.0277	0.9320 ± 0.0145
AGNews	`random_edge_dropout`	0.9231 ± 0.0123	0.8507 ± 0.0255	0.9974 ± 0.0048	0.8813 ± 0.0196	0.9441 ± 0.0117
AGNews	`random_full_budget`	0.8578 ± 0.0222	0.7387 ± 0.0362	1.0000 ± 0.0000	0.8169 ± 0.0274	0.9303 ± 0.0139
CNN_DM	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN_DM	full	0.8308 ± 0.0144	0.7003 ± 0.0239	1.0000 ± 0.0000	0.7749 ± 0.0200	0.9028 ± 0.0132
CNN_DM	`random_edge_dropout`	0.9157 ± 0.0149	0.8400 ± 0.0276	0.9993 ± 0.0014	0.8671 ± 0.0262	0.9347 ± 0.0170
CNN_DM	`random_full_budget`	0.8258 ± 0.0186	0.6940 ± 0.0299	1.0000 ± 0.0000	0.7658 ± 0.0294	0.8966 ± 0.0190
IMDB	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
IMDB	full	0.8358 ± 0.0223	0.7029 ± 0.0350	1.0000 ± 0.0000	0.7783 ± 0.0316	0.8950 ± 0.0172
IMDB	`random_edge_dropout`	0.9218 ± 0.0204	0.8482 ± 0.0378	0.9963 ± 0.0070	0.8745 ± 0.0347	0.9251 ± 0.0252
IMDB	`random_full_budget`	0.8532 ± 0.0252	0.7341 ± 0.0397	1.0000 ± 0.0000	0.8012 ± 0.0346	0.9080 ± 0.0225
PubMed	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
PubMed	full	0.8317 ± 0.0221	0.7089 ± 0.0326	1.0000 ± 0.0000	0.8670 ± 0.0199	0.9925 ± 0.0051
PubMed	`random_edge_dropout`	0.9198 ± 0.0141	0.8468 ± 0.0252	0.9991 ± 0.0014	0.9142 ± 0.0168	0.9808 ± 0.0071
PubMed	`random_full_budget`	0.8316 ± 0.0184	0.7053 ± 0.0255	1.0000 ± 0.0000	0.8644 ± 0.0210	0.9913 ± 0.0065
SynthPII	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
SynthPII	full	0.8136 ± 0.0185	0.6724 ± 0.0295	1.0000 ± 0.0000	0.7111 ± 0.0287	0.7580 ± 0.0204
SynthPII	`random_edge_dropout`	0.9127 ± 0.0172	0.8384 ± 0.0277	0.9974 ± 0.0063	0.8485 ± 0.0278	0.8613 ± 0.0273
SynthPII	`random_full_budget`	0.8250 ± 0.0218	0.6881 ± 0.0319	1.0000 ± 0.0000	0.7263 ± 0.0327	0.7685 ± 0.0292

Table 8: Relevant-subgraph structural utility. Recall measures retention of original evidence, whereas Jaccard similarity additionally accounts for replacement structure.

Dataset	Variant	Relevant-node recall	Relevant-node Jaccard	Relevant-edge recall	Relevant-edge Jaccard
AGNews	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
AGNews	full	0.8424 ± 0.0240	0.7079 ± 0.0351	0.7379 ± 0.0253	0.6177 ± 0.0311
AGNews	`random_edge_dropout`	0.8943 ± 0.0142	0.8943 ± 0.0142	0.8257 ± 0.0164	0.8257 ± 0.0164
AGNews	`random_full_budget`	0.8340 ± 0.0246	0.6965 ± 0.0353	0.7290 ± 0.0279	0.6076 ± 0.0343
CNN_DM	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN_DM	full	0.7890 ± 0.0144	0.6101 ± 0.0218	0.6724 ± 0.0156	0.5083 ± 0.0224
CNN_DM	`random_edge_dropout`	0.8830 ± 0.0216	0.8830 ± 0.0216	0.8170 ± 0.0224	0.8170 ± 0.0224
CNN_DM	`random_full_budget`	0.7845 ± 0.0207	0.6080 ± 0.0219	0.6635 ± 0.0217	0.5014 ± 0.0200
IMDB	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
IMDB	full	0.8003 ± 0.0280	0.6695 ± 0.0361	0.7105 ± 0.0299	0.5456 ± 0.0324
IMDB	`random_edge_dropout`	0.8882 ± 0.0267	0.8882 ± 0.0267	0.8292 ± 0.0296	0.8292 ± 0.0296
IMDB	`random_full_budget`	0.8142 ± 0.0312	0.6836 ± 0.0342	0.7234 ± 0.0331	0.5554 ± 0.0317
PubMed	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
PubMed	full	0.8659 ± 0.0199	0.7863 ± 0.0250	0.7219 ± 0.0211	0.6046 ± 0.0234
PubMed	`random_edge_dropout`	0.9273 ± 0.0121	0.9273 ± 0.0121	0.8520 ± 0.0117	0.8520 ± 0.0117
PubMed	`random_full_budget`	0.8659 ± 0.0161	0.7864 ± 0.0214	0.7204 ± 0.0168	0.6026 ± 0.0184
SynthPII	baseline	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
SynthPII	full	0.7230 ± 0.0253	0.5941 ± 0.0290	0.6649 ± 0.0255	0.5295 ± 0.0287
SynthPII	`random_edge_dropout`	0.8601 ± 0.0188	0.8601 ± 0.0188	0.8274 ± 0.0188	0.8274 ± 0.0188
SynthPII	`random_full_budget`	0.7350 ± 0.0187	0.6029 ± 0.0217	0.6765 ± 0.0191	0.5378 ± 0.0212

6.5. End-to-End Relation-Question-Answering Utility and Sensitive-Relation Disclosure

The locked end-to-end evaluation completed all 43,080 planned model generations over 718 fixed queries, five datasets, 15 protection seeds, and four graph conditions. The execution audit found no duplicate rows, query-set drift, or determinism mismatches, and confirmed successful degree-preservation and 2K joint-degree checks (Supplementary Table S26). The primary analysis retained all 26 parse failures, corresponding to 0.06035% of generated records.

For non-sensitive relation questions, full TAFP-GraphRAG achieved relation-token F1 of 0.640579 versus 0.682930 for the unprotected baseline (Table 9), retaining 93.8% of baseline F1. The paired difference was -0.042351, with a 95% interval of [-0.061646, -0.023055] and Holm-adjusted p = 0.000610. Retrieved-evidence recall decreased from 0.976940 to 0.874470. In contrast, exact sequence match increased from 0.161537 to 0.184574 and exact faithfulness to the retrieved relation sequence from 0.181931 to 0.306207; both paired differences were significant after Holm correction. The result therefore reflects a bounded utility trade-off rather than lossless preservation.

For sensitive-relation questions, disclosure decreased from 0.970000 under the baseline to 0.075600 under full TAFP-GraphRAG. The absolute reduction was 0.894400, corresponding to a 92.2% relative reduction. Its 95% interval was [0.890469, 0.898331], with Holm-adjusted p = 0.001863. Sensitive-token recall decreased from 0.932000 to 0.062978 and retrieved sensitive-evidence exposure from 0.942000 to 0.049311.

The sensitivity-guided condition also separated from both sensitivity-blind controls downstream. Full TAFP-GraphRAG achieved relation-token F1 values 0.502270 and 0.501185 higher than `random_full_budget` and 2K, respectively; both Holm-adjusted p values were 0.000183. Its sensitive-disclosure rate was 0.156933 lower than `random_full_budget` and 0.169067 lower than 2K, with Holm-adjusted p = 0.001863 for both comparisons. These directions and adjusted significance results held for the two primary outcomes in each of the five datasets and overall (Supplementary Table S25).

Performance remained dataset dependent. Under full protection, relation-token F1 ranged from 0.587031 to 0.706349 and sensitive-disclosure rate from 0.000000 to 0.155333 (Supplementary Table S24). Disclosure was zero on PubMed and 0.011333 on SynthPII; residual disclosure remained on CNN/DM, AGNews, and IMDB, reaching 0.155333 on IMDB. The parse-success-only sensitivity analysis left the principal full and baseline F1 and disclosure values unchanged.

Together with Section 6.4, these findings refine the empirical contribution. The method is not supported as a universal mechanism for maximizing structural overlap because the matched random full-budget control was often structurally similar. Under the predefined end-to-end relation-QA protocol, however, sensitivity-guided transformation preserved substantially more non-sensitive answering utility while disclosing fewer sensitive relations than both evaluated sensitivity-blind controls. This supports selective, task-grounded relation protection under the evaluated conditions; it does not establish formal privacy, complete concealment, raw-text privacy, or protection against all attackers.

Figure 7 summarizes the end-to-end utility and sensitive-relation exposure outcomes.

Table 9: Retrieval-grounded end-to-end relation-QA results across five datasets. Values are means over 15 protection seeds. Higher values are better for relation-token F1, exact match, and retrieved-evidence recall; lower values are better for the three disclosure metrics. The locked protocol comprised 43,080 generations.

Variant	Relation-token F1	Exact match	Retrieved-evidence recall	Sensitive-relation disclosure	Sensitive-token recall	Retrieved sensitive exposure
Unprotected baseline	0.6829	0.1615	0.9769	0.9700	0.9320	0.9420
Full TAFP	0.6406	0.1846	0.8745	0.0756	0.0630	0.0493
Random full-budget	0.1383	0.0223	0.1596	0.2325	0.1735	0.1553
2K	0.1394	0.0238	0.1600	0.2447	0.1816	0.1643

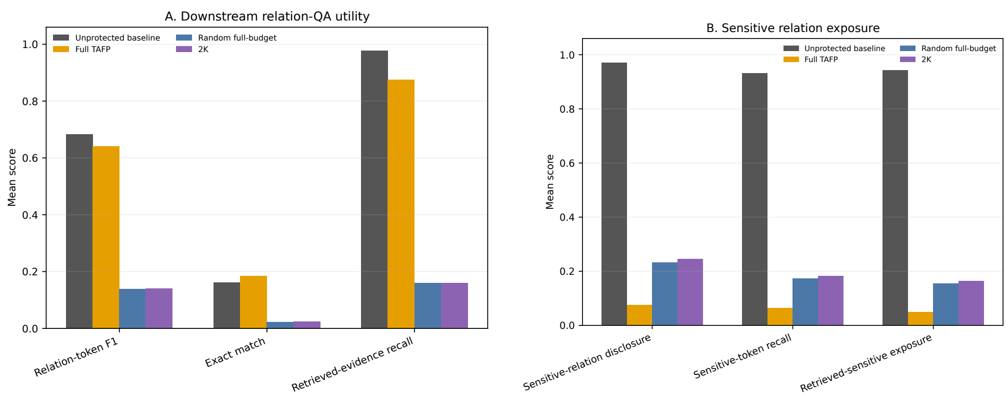

Figure 7: End-to-end retrieval-grounded relation-QA utility and sensitive-relation exposure. Panel A reports relation-token F1, exact match, and retrieved-evidence recall for the unprotected baseline, Full TAFP, the random full-budget control, and the 2K baseline. Panel B reports sensitive-relation disclosure, sensitive-token recall, and retrieved-sensitive exposure, where lower values indicate lower downstream exposure. Values are aggregate results from the locked 43,080-generation evaluation protocol.

7. Discussion

7.1. Policy Guidance Determines Where Information Loss Occurs

The central empirical contribution is the policy-conditioned allocation of perturbation, not perturbation alone. Full TAFP-GraphRAG concealed 0.99993–1.00000 of policy-designated sensitive relation labels while retaining 0.89935–0.92193 of non-sensitive relation semantics. The perturbation-budget-matched random control retained nearly the same fraction of original edges, yet it concealed fewer sensitive labels and retained substantially less non-sensitive semantics. The minority-sensitive ablation reproduced the same concealment–retention profile from 5% to 30% sensitive-edge prevalence. Together, these results identify sensitivity guidance as the mechanism determining which relation semantics are removed and retained.

This interpretation is narrower than a claim of universal optimization. At 5% sensitive prevalence, the random relation-budget control retained more non-sensitive semantics, but it concealed only 0.0475 of sensitive labels, compared with 0.9996 under full protection. The relevant contribution is therefore selective information allocation under an explicit policy, not maximization of any single utility metric independently of the protection objective.

The comparative pattern depended on the evaluation objective. Policy guidance produced its largest gains when success required concealing designated labels while preserving the original semantics of non-sensitive edges, and it also separated from both sensitivity-blind controls in end-to-end relation QA. Matched random perturbation was more competitive for structural-overlap metrics and for several topology- or representation-based attackers under the same perturbation budget. Sensitive prevalence, label composition, and attacker observation surface therefore moderated the observed advantage; graph density and label entropy were not isolated as causal factors in the present design.

7.2. Relation Privacy Must Be Evaluated Across Distinct Attack Surfaces

The results distinguish relation-label concealment, policy-membership privacy, original-edge privacy, and representation-based relation inference as separate security properties. Direct recovery of policy-designated sensitive relation labels was eliminated under the evaluated relation-only and full protocols. This did not remove evidence that an original edge had existed: hybrid edge-inference AUC remained high on several datasets. Topology-only, spectral, and released-view GCN-GAE attacks were also dataset dependent.

PubMed showed significant reductions relative to the unprotected graph, but the matched random full-budget control was statistically competitive; SynthPII did not show a consistent reduction.

The marker audit identified an additional release-layer risk. The default PRIVATE_RELATION marker and relation-label omission both revealed supplied-policy membership perfectly in the evaluated protocol. Coarse shared markers reduced that leakage only by sacrificing non-sensitive semantic retention. A protected graph therefore cannot be assessed solely by checking whether the original sensitive label is absent. The released schema, marker vocabulary, retained topology, and learned graph representations each define a separate observation surface and require separate measurements.

7.3. Structural Invariants Constrain Transformation but Do Not Establish Privacy

Exact per-node degree preservation was satisfied in every designated degree-preserving condition, and the 2K control additionally preserved the complete joint-degree distribution. These strong, directly auditable structural invariants do not imply relation privacy or semantic selectivity. The 2K control did not reproduce the concealment and non-sensitive retention achieved by the sensitivity-guided method, and identical degree constraints did not prevent residual topology- or representation-based inference.

Structural invariants therefore bound graph change and support compatibility and reproducibility, while the sensitivity policy determines where semantic protection is applied. Neither role is equivalent to a formal differential-privacy guarantee. TAFP-GraphRAG should accordingly be interpreted as an empirical, policy-conditioned protection mechanism with verified structural constraints, not as a substitute for differential privacy.

Exact degree preservation can also retain distinctive degree signatures that may act as quasi-identifiers or auxiliary inference features. The study therefore treats the invariant as a compatibility constraint rather than a privacy objective.

7.4. Utility Conclusions Depend on the Evaluation Layer

The structural evaluation showed substantial retrieval-support preservation: all evaluated endpoint pairs remained reachable, most path-support edges were retained, and most relevant nodes and edges remained available. Nevertheless, the matched random full-budget control was structurally similar on most path and subgraph comparisons, and random edge dropout preserved more exact original structure on most measures. These findings do not support a general claim that sensitivity guidance maximizes structural overlap.

The end-to-end relation-QA results provide a different and more task-grounded comparison. Full TAFP-GraphRAG retained 93.8% of baseline non-sensitive relation-token F1 and reduced sensitive-relation disclosure by 92.2% relative to the baseline. It also achieved higher non-sensitive relation-token F1 and lower sensitive disclosure than both evaluated sensitivity-blind controls in the overall analysis and in every dataset. At the same time, retrieval recall and relation-token F1 remained below the unprotected baseline, whereas exact sequence match and exact faithfulness increased. These directions show why utility must be characterized jointly rather than reduced to a single structural or generation score.

Together, the results support a precise downstream claim: sensitivity guidance did not generally preserve more graph structure than an equal-budget random transformation, but under the predefined relation-QA protocol it preserved substantially more task-relevant non-sensitive answering utility while disclosing fewer sensitive relations than the evaluated sensitivity-blind controls.

7.5. Policy Quality and Release Design Are Operational Dependencies

The policy-misclassification experiment showed that TAFP-GraphRAG faithfully executes the supplied policy but does not repair it. False negatives reduced concealment measured against the oracle policy, while false positives reduced oracle non-sensitive preservation. This distinguishes transformation correctness from policy correctness. In deployment, sensitivity-policy validation, versioning, and error auditing are therefore part of the protection process rather than external administrative details.

Release design requires equal attention. A marker that conceals the original relation label may still expose whether an edge was designated sensitive. Eliminating this policy-membership signal through broader cover labels produced a direct semantic-utility cost in the evaluated settings. Operational use therefore requires an explicit decision about which disclosure object is protected: the original relation value, the fact of policy designation, the existence of an original edge, or some combination of these.

No evaluated shared-cover setting was universally optimal. Applying the cover to all non-sensitive relations reduced policy-membership AUC to chance but eliminated measured non-sensitive semantic retention, whereas partial covers produced intermediate leakage and utility.

7.6. Implications for Security-Oriented GraphRAG Deployment

The findings support an offline governance pattern in which policy selection, graph transformation, invariant checks, attack-specific evaluation, and hash-traceable audit records are completed before the protected graph is used for retrieval. This pattern exposes residual risks rather than treating the transformation name as evidence of privacy, and allows deployment decisions to reflect the threat model and acceptable trade-offs between non-sensitive utility and the disclosure objects being protected.

Within the evaluated scope, TAFP-GraphRAG contributes a reproducible method for selective relation protection that combines sensitivity-aware semantic transformation, designated structural invariants, and attack-specific privacy–utility evaluation. Its strongest evidence concerns selective relation-semantic protection and downstream relation-QA behavior. Attack resistance remains dataset and adversary dependent, and the framework does not establish formal privacy, complete anonymization, raw-text privacy, or protection against adaptive multi-turn reconstruction.

8. Limitations and Scope Boundaries

The evidence supports selective protection of policy-designated relation semantics under the evaluated protocols, but it does not establish universal or formal privacy. TAFP-GraphRAG does not provide differential privacy or local differential privacy, and the reported structural invariants must not be interpreted as equivalent guarantees. The framework also does not address raw-text confidentiality, document-membership privacy, or complete anonymization.

Attack resistance was dataset and attacker dependent. Direct recovery of designated sensitive relation labels was eliminated in the evaluated relation-level protocols, but residual inference remained through other observation surfaces. Original-edge existence could remain inferable, and the strongest released-view GCN-GAE stress test retained non-trivial predictive performance on PubMed. In several attack settings, the matched `random_full_budget` control was statistically competitive. The evaluated attacks therefore bound the demonstrated protection; they do not exhaust adaptive, coordinated, or future adversaries.

The attack coverage was not uniform across all datasets. The structure-only, spectral, and released-view GCN-GAE evaluations were conducted on

PubMed and SynthPII, whereas the fixed-edge, semantic-selectivity, marker-leakage, policy-misclassification, structural-utility, and end-to-end protocols covered five datasets. The reported attack conclusions should therefore remain tied to their corresponding datasets and observation models.

The sensitivity policies derived for the public datasets are experimental proxy policies rather than validated enterprise confidentiality labels. The policy-misclassification results show that the transformation executes the supplied policy but does not correct it. False negatives can leave oracle-sensitive relations insufficiently protected, and false positives can remove non-sensitive semantics unnecessarily. Real deployments therefore require domain-specific policy construction, validation, version control, and error auditing before transformation.

The default protected marker did not conceal whether an edge belonged to the supplied sensitive-policy set. Shared cover labels reduced this policy-membership signal only with a measurable reduction in non-sensitive semantic retention. Consequently, the present framework does not simultaneously optimize original-label concealment, policy-membership concealment, edge-existence privacy, and semantic utility. Deployment must specify which disclosure objects are in scope.

Deployments should restrict released edge attributes through an explicit allowlist. This recommendation defines a deployment boundary and was not separately evaluated in the present experiments.

Utility preservation was substantial but not lossless. Full protection reduced non-sensitive relation-token F1 and retrieved-evidence recall relative to the unprotected baseline, even though it improved exact sequence match and exact faithfulness and outperformed both evaluated sensitivity-blind controls on the primary downstream outcomes. Structural overlap also was not generally higher than under the matched random full-budget control. The results therefore support a task-conditioned privacy–utility trade-off, not universal utility preservation.

The end-to-end findings are bounded to the locked model, fixed query set, datasets, graph-construction rules, sensitivity policies, and four evaluated graph conditions. The study did not evaluate adaptive multi-turn reconstruction, collusion across releases, continual graph updates, cross-domain enterprise deployment, or long-term policy drift. These extensions are important future evaluation directions, but their outcomes cannot be inferred from the present evidence.

Computational evidence is limited to apply_variant runtime and the controlled 2K telemetry on the five evaluated graphs. Peak memory, proposal-level attempt counts, proposal-acceptance ratios, and empirical scaling beyond these graph sizes were not instrumented. The study therefore makes no claim about production throughput or asymptotic scalability.

9. Conclusion

This study introduced TAFP-GraphRAG as an empirical, policy-conditioned framework for protecting sensitive relation semantics in GraphRAG while preserving explicitly defined structural invariants and maintaining auditable transformation records. Across five datasets, the full configuration concealed 0.99993–1.00000 of policy-designated sensitive relation labels, retained 0.89935–0.92193 of non-sensitive relation semantics, and preserved the exact degree of every node in the designated degree-preserving conditions. Equal-budget random and 2K-constrained controls did not reproduce this semantic selectivity, showing that the observed protection profile depended on sensitivity-guided allocation rather than perturbation volume or degree constraints alone.

The attack evaluation also established the boundaries of this contribution. Direct recovery of designated sensitive relation labels was eliminated in the evaluated relation-level protocols, but original-edge existence and representation-based relation inference could remain possible. Attack reductions were dataset and attacker dependent, the matched random full-budget control was competitive in several attack settings, and the default protected marker revealed supplied-policy membership. These findings support attack-specific evaluation of protected GraphRAG releases rather than a single undifferentiated privacy score.

Utility results were likewise evaluation-layer dependent. Full TAFP-GraphRAG preserved substantial path and local-subgraph support but did not generally exceed the matched random full-budget control in structural overlap. However, under the predefined retrieval-grounded structured relation-sequence question-answering protocol with deterministic decoding, it retained 93.8% of baseline non-sensitive relation-token F1 and reduced sensitive-relation disclosure by 92.2% relative to the unprotected baseline. It also achieved higher non-sensitive relation-token F1 and lower sensitive disclosure than both evaluated sensitivity-blind controls overall and in every dataset.

The resulting contribution is therefore precise: under the evaluated policies, datasets, models, and threat models, TAFP-GraphRAG provides reproducible selective protection of relation semantics with exact degree preservation in designated variants and task-grounded privacy–utility benefits. It does not provide differential privacy, complete anonymization, raw-text privacy, edge privacy, topology privacy, or universal attack resistance. Future work should evaluate alternative release markers, validated domain policies, adaptive and cross-release attacks, continual graph updates, and real enterprise deployments without extending the present evidence beyond its verified scope.

Supplementary Materials

The supplementary materials provide detailed evidence supporting the main findings, including variant-level integrity audits, conditional relation-recovery results, feature-ablation analyses, fixed-edge and multi-split attack details, path- and subgraph-level paired tests, semantic-selectivity comparisons, sensitive-relation schema documentation (Tables S1–S18), sensitivity-flag leakage and cover-set coarsening (Table S20), sensitivity-policy misclassification robustness (Table S21), spectral embedding-based relation-inference summaries and paired tests (Tables S22–S23), dataset-level end-to-end relation-QA results (Table S24), paired seed-level end-to-end comparisons (Table S25), the parse-failure and execution audit (Table S26), the minority-sensitive policy ablation (Table S27), and the released-view GCN-GAE summary and paired tests (Tables S28–S29). Parameter-sensitivity protocols are described in Section 5.2 and their verified evidence is retained in the supplementary and immutable experiment archives. Distributed evidence tables and audit records are linked through hash-verified manifests. A claim-to-test evidence map linking the principal manuscript claims to their primary outcomes, comparators, experimental scopes, inferential units, multiplicity procedures, and evidence locations is provided as a separate supplementary index.

CRediT authorship contribution statement

Yılmaz Vural: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review and editing, Visualization, Project administration.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics statement

This study used publicly available benchmark datasets and synthetically generated data. It did not involve human participants, identifiable private human-subject data, clinical intervention, or animal subjects. Institutional review board approval and informed consent were therefore not applicable.

Data availability

The third-party benchmark datasets used in this study are publicly available from their original providers under their respective licenses; the study also used synthetically generated data. The manuscript source, supplementary tables, claim-to-test evidence map, audit records, and SHA-256 manifests supporting the reported analyses are publicly available in the TAFP-GraphRAG submission-support repository. Third-party benchmark datasets are not redistributed. Additional processed or synthetic artifacts can be provided to the editor and reviewers upon reasonable request, subject to applicable licenses and platform restrictions.

Code availability

The public submission-support repository contains the complete LaTeX source, figures, supplementary materials, audit records, and manifests for the submitted study, but it is not a complete public release of the implementation and evaluation code. The implementation and evaluation scripts are maintained in an access-controlled project repository and can be provided to the editor and reviewers upon reasonable request, subject to licensing, security, and repository-governance restrictions.

AI-assisted figure declaration

The scientific figures in this manuscript were not generated using generative AI. They were produced programmatically from verified source data and documented scientific specifications using TikZ and Matplotlib. All figure contents, labels, numerical values, and layouts were reviewed by the author.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to support language refinement, document organization, consistency checking, and formatting review. The tool was not treated as an author and did not independently determine the scientific claims, experimental design, results, or conclusions. The author reviewed and edited all AI-assisted material and takes full responsibility for the content of the article.

Acknowledgments

The author acknowledges the computing environments used to execute and validate the experiments.

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